Convex Polytopes and Toric Topology

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Contents

1 Introduction

The project intends to study some interactions between convex geometry, topology and combinatorics. To achieve this, the project consists of three main aims:

(1) Introduce the theory of convex geometry with an emphasis on convex polytopes. Convex polytopes are special convex sets in $\mathbb{R}^n$ that can be defined by two equivalent ways: an intersection of a finite number of closed half-spaces that is bounded(so called H-polytopes), or the convex hull of finitely many points called vertices(so-called V-polytopes). The definition of H-polytopes gives explicit inequalities describing half-spaces so is useful for linear algorithms involving polytopes, for example in linear programming, while the definition of V-polytopes is more geometric and is used to analysis their topological properties. They both have a close connection with the combinatorial information carried by polytopes and are used extensively in various branches of convex geometry and applications in combinatorics and topology etc. Therefore, the equivalence between H-polytopes and V-polytopes is seen as one of the most fundamental results in convex geometry, which however is not easy to prove. In this project, we aim to give a proof of this equivalence by developing topological properties of convex sets and the polarity of convex polytopes.

(2) Introduce simplicial complexes as an example of bridges between convex geometry, topology and combinatorics. Simplicial complexes are a certain set of simplices(which are convex polytopes) satisfying some conditions. It is often used in algebraic topology for concrete calculations of algebraic attributes like homology and cohomology, but the notion itself is combinatorial. In this project, we aim to discuss two typical examples: simplicial decomposition of convex polytopes and the combinatorial Alexander duality. Each of them reflects the connection to convex geometry and topology respectively.

(3) Discuss a topological construction of toric varieties built upon convex polytopes and calculate the homology groups of all toric varieties of polygons. Toric varieties are important geometric objects in algebraic geometry which are fully specified by combinatorial data they carry. Based on the knowledge of convex polytopes we built up in the first part, we aim to understand the relation between toric varieties and convex polytopes: every convex polytope P can be associated with a toric variety by an algebraic construction as we will see. Furthermore, this toric variety has an equivalent topological description, namely it can be identified with the product of the polytope with an n-torus under some equivalence relations. Our primary hope is to understand the homotopy type of certain toric varieties(with the above topological description) by calculating their homology groups. In particular, we will discuss the homology groups of toric varieties associated with polygons in detail.

Section 2 of the project is devoted to convex geometry, which mainly studies the subsets of $\mathbb{R}^n$(or any real or complex vector space) with the property that the segment connecting any two points in the set is also contained in the set. In low-dimensional cases($n \leq 3$), many geometric properties of convex sets and polytopes are rather "obvious" in the sense that intuition usually leads to correct answer. However, most of them do require rigorous proofs and some theorems are true in low-dimensions but fail in higher dimensions. We will discuss various properties and results of convex sets of $\mathbb{R}^n$. Most of them are valid in any real or complex vector spaces. After that, we move to a detailed treatment of convex polytopes. We will classify all faces of a convex polytope, and discuss the polarity of polytopes. Simplicial and simple polytopes are also introduced. The main aim is to give a proof of the equivalence of an H-polytope and a V-polytope. In addition, this section also provides sufficient background knowledge for section 3 and 4.

Next, in section 3, we discuss a version of the Alexander duality for simplicial complexes. The Alexander duality is originally proved by J.W. Alexander and plays an important role in the topology of manifolds. For example, it can be used to prove that $\mathbb{R}P^2$ can not be embedded in $\mathbb{R}^3$. We will discuss a special case of the Alexander duality: the combinatorial Alexander duality, which is a nice illustration of the connections between combinatorics and topology. Although we will not touch the theory, it is used in the study of moment-angle complexes. Furthermore, we will also draw the connections between convex geometry and simplicial complexes by discussing the simplicial decomposition of convex polytopes: every convex polytope has a simplicial decomposition.

Finally, in section 4, we discuss a topological construction of toric varieties from convex polytopes. We will start from a short discussion of classical definitions from polyhedral cones and fan language with some low-dimensional examples like $\mathbb{CP}^2$. Our aim, however, is to discuss an equivalent, topological formulation of the toric variety of the normal fan of a convex polytope P and calculate their homology groups. In particular, every n-polytope has a normal fan corresponds to its facets($n-1$-dimensional faces) which can be associated with a toric variety. This variety is identified with a quotient space $P \times T^n / \sim$ for some equivalent relation depends on the normal fan (where T^n stand for the n-torus). We will calculate the homology groups of all toric varieties constructed from 2-dimensional convex polytopes by using elementary tools: the Mayer-Vietories sequences and the Künneth formula. We will work out an important example in detail: the Hirzebruch surface, which is defined based on the combinatorial data of a quadrilateral. This also provides a general strategy for higher dimensions cases, although the amount of calculation might become an obstruction.

2 Convex geometry and polytopes

Convex geometry is the study of convex sets in $\mathbb{R}^n$. The aim of section is to give an self-contained introduction to convex geometry with an emphasis on convex polytopes. The material in this section is not only interesting as a separate subject itself, but also appears as a language in many other areas of mathematics: combinatorics, algebraic geometry and topology. This section also services as background knowledge for our discussion of toric topology in section 4.

We start from establishing the language of convex sets and then turn to the theory of convex polytopes and introduce their basic properties. There are two different definitions of convex polytopes which are equally useful. Our aim is to give a proof of the main theorem of convex polytopes, namely that are actually equivalent. We will develop the proof through the topological properties of convex subsets in $\mathbb{R}^n$ and the notion of dual polytopes. Most of the material in this section can be found in [2] and [8].

2.1 Affine spaces and convex sets

Definition 2.1. A subset X of $\mathbb{R}^n$ is **convex** if for any points $a, b \in X$, the line segment $\{\lambda a + (1-\lambda)b \colon 0 \leq \lambda \leq 1\}$ is also contained in X.

Definition 2.2. Let X be a subset of $\mathbb{R}^n$. The **convex hull** of X, denoted by $\operatorname{conv}(X)$, is the intersection of all convex subsets of $\mathbb{R}^n$ containing X.

For any finite subset $\{x_1, \dots, x_k\} \subseteq X$ and non-negative coefficients $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ with $\sum_{i=1}^k \lambda_i = 1$, we have $\lambda_1 x_1 + \cdots + \lambda_k x_k = (1-\lambda_k)(\frac{\lambda_1}{1-\lambda_k} x_1 + \cdots + \frac{\lambda_{k-1}}{1-\lambda_k} x_{k-1}) + \lambda_k x_k$. So by a simple induction, any such combination $\lambda_1 x_1 + \cdots + \lambda_k x_k$ must be contained in $\operatorname{conv}(X)$. Therefore, we can formulate the convex hull of X as

$$\operatorname{conv}(X) = \{\lambda_1 x_1 + \cdots + \lambda_k x_k \colon \{x_1, \dots, x_k\} \subseteq X \; \lambda_i \geq 0, \sum_1^k \lambda_i = 1\}$$

and any expression in the form of $\lambda_1 x_1 + \cdots + \lambda_k x_k$ is called a **convex combination**. X is sometimes called a generating set.

Definition 2.3. Let V be a vector space over a field k. For $a_i \in k$ and $v_i \in V$, $i = 1, \dots, n$, the linear combination $a_1 v_1 + \cdots + a_n v_n$ is called an **affine combination** if $\sum_1^n a_i = 1$. A subset $U \subseteq V$ is called an **affine subset** or **affine subspace** if the affine combinations of any finite subset of vectors in U are also contained in U.

Definition 2.4. Let X be a subset of some $\mathbb{R}^n$. The **affine hull** of X, denoted by $\operatorname{aff}(X)$, is the intersection of all affine subsets of $\mathbb{R}^n$ containing X.

Similar to convex hull, we can formulate the affine hull of X as $\operatorname{aff}(X) = \{\lambda_1 x_1 + \cdots + \lambda_k x_k \colon \{x_1, \dots, x_k\} \subseteq U, \lambda_i \in \mathbb{R}, \sum_1^k \lambda_i = 1\}$. The following simple proposition summarises the basic properties of affine subsets.

Proposition 2.5 (Based on section 2.1 of [8])**.** *Let V be a vector space and U be a subset of V, then the following conditions are equivalent:*

1. *U is an affine set.*

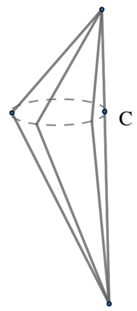


Figure 1: The space X

2. *$\alpha x + (1-\alpha)y$, $x+y-z \in U$ for all $x, y, z \in U$ and all $\alpha \in k$.*

3. *For all $a \in U$, $U - a = \{u - a : u \in U\}$ is a linear subspace.*

4. *There exists a linear subspace W and a vector $b \in V$ such that $U = W + b = \{w + b : w \in W\}$.*

Proof. $1 \Rightarrow 2$ Follows from the definition, since $\alpha x + (1-\alpha)y$, $x+y-z \in U$ are affine combinations.

$2 \Rightarrow 3$ Since $a \in U$, we have $0 = a - a \in U - a$. Now let $r, s \in U - a$ and $\alpha \in k$, we have $r + \alpha s + a = \alpha(s+a) + (r+a) - (\alpha - 1)a \in U$. Hence $r + \alpha s \in U - a$.

$3 \Rightarrow 4$ Let b be any element of U and let $W = U - b$.

$4 \Rightarrow 1$ Let $u_1, \ldots, u_k \in U$ and $\sum_1^k \alpha_i = 1$ for $\alpha_i \in k$. We have $\alpha_1 u_1 + \cdots + \alpha_k u_k - b = \alpha_1(u_1 - b) + \cdots + \alpha_k(u_k - b) \in W$. Hence $\alpha_1 u_1 + \cdots + \alpha_k u_k \in W + b = U$. $\square$

Thus affine subsets of a vector space can be seen as translations of linear subspaces, so we also define the **dimension** of an affine set to be the dimension of its corresponding linear subspace. We are mainly working with affine subspaces of $\mathbb{R}^n$. Let's now turn to some topological properties of convex subsets of affine subspaces and convex sets of $\mathbb{R}^n$.

Definition 2.6. Let $U \subset \mathbb{R}^n$ be an affine space and $A \subset U$ be a convex subset. A point $a \in \partial A$, the topological boundary of A in U, is called an **extreme point** if $A \setminus \{a\}$ is still a convex subset.

Example 2.7. The cube $I^3 = \{(x,y,z) \in \mathbb{R}^3 : (x^2+y^2+z^2)^{1/2} \leq 1\}$ in $\mathbb{R}^3$ has extreme points $(\pm 1, \pm 1, \pm 1)$ which are the vertices. We will see later that the extreme points of any convex polytope coincide with the vertices.

It is useful to note that the set of extreme points of a convex subset is not guaranteed to be a closed set in $\mathbb{R}^n$. For example, consider the set $X = \operatorname{conv}(\{(x,y,0) : x^2+y^2 = 1\} \cup \{(1,0,\pm 1)\})$ as figure 1 shows. The extreme points of X are $E = \{(x,y,0) : x^2+y^2 = 1, x \neq 1\} \cup \{(1,0,\pm 1)\}$. In particular, the point C is not an extreme point so E is not closed.

Lemma 2.8. *(lemma 2.7 of [10]) Let $U \subset \mathbb{R}^n$ be an n-dimensional affine space, and $A \subseteq U$ be a convex and compact subset. Then A is the convex hull of its boundary ∂A.*

Proof. Let $x \in \text{int}A$, the intersection $B = l \cap A$ between a line l through x and A is a closed, convex set since l is closed and convex. Since A is compact, B is also compact. Therefore, the only possibility of B is a closed line segment from a to b containing x with some $a, b \in \partial A$. a, b are boundary points since A is homeomorphic to some closed ball.(see for example Stefan's notes, [9]). Hence a is in the convex hull of ∂A. $\square$

This lemma also coincides with our intuition. Actually we have better:

Theorem 2.9. *(Theorem 2.8 of [10]) Let $U \subset \mathbb{R}^n$ be an n-dimensional affine space, and $A \subseteq U$ is a convex and compact subset. Then A is the convex hull of its extreme points.*

The proof of this theorem is more subtle and we need a few more results before tackling it.

Lemma 2.10. *(Lemma 3.1 of [10]) Let $X \subset \mathbb{R}^n$ be a non-empty convex set with two points $a \in$ int(X) and $b \in \overline{X}$(closure of X). Then the open line segment l, defined by $l = \{(1-\lambda)a + \lambda b \colon \lambda \in (0,1)\}$ is contained in int(X).*

Proof. This intuitively clear. To give a proof, for each $z \in l$ and the coefficient corresponding λ, consider the homeomorphism $f_z(t) = z + \frac{\lambda-1}{\lambda}t$. f_z maps a to b. $\text{int}(X)$ is open, so there exists some open neighbourhood of $a \in U \subset \text{int}(X)$. $V = f_z(U)$ is an open neighbourhood of b with $V \cap X \neq \emptyset$ since $b \in \overline{X}$. Let $v \in V \cap X$ and $u = f_z^{-1}(v)$, and we have $z = (1-\lambda)u + \lambda v \in X$. Consider the open sub-set $W = \{(1-\lambda)w + \lambda v \colon w \in U\}$. W is open since it is obtained from U by expansion and translation. It's clear that $z \in W$, hence $z \in \text{int}(X)$. $\square$

Corollary 2.11. *(Corollary 3.2 of [10]) Let $X \subset \mathbb{R}^n$ be a convex set with non-empty interior, then*

1. *int(X) is also convex*
2. *int(X) =int$(\overline{X})$*
3. *$\overline{X} = \overline{\text{int}(X)}$*

Proof. 1 follows easily from lemma 2.10. For 2, we need to show $\text{int}(\overline{X}) \subset \text{int}(X)$. Let $x \in \text{int}(\overline{X})$, there exists some open ball B_r with positive radius in $\overline{X}$ containing x. Choose any open line segment l in B_r containing x, we have $x \in \text{int}(X)$ by lemma 2.10. For 3, we need to show $\overline{X} \subset \overline{\text{int}(X)}$. Let $x \in \overline{X}$ and $y \in \text{int}(X)$, the open line segment l is contained in $\text{int}(X)$ so the result follows from taking closure. $\square$

Definition 2.12. The $n+1$ points $v_1, \ldots, v_{n+1}$ in $\mathbb{R}^n$ are **affinely independent** if their affine hull has dimension n.

Lemma 2.13. *(Based on he brief discussion in chapter 0 of [8]) The following conditions are equivalent:*

1. *$v_1, \dots, v_{n+1} \in \mathbb{R}^n$ are affinely independent.*
2. *For all $\lambda_i \in \mathbb{R}$ with $\sum_1^{n+1} \lambda_i = 0$, $\sum_1^{n+1} \lambda_i v_i = 0$ implies $\lambda_i = 0$.*

Proof. Suppose $v_1, \dots, v_{n+1}$ are affinely independent. Let $u_i = v_i - v_{n+1}$ for $i = 1, \dots, n+1$, then $u_{n+1} = 0$. We have $\sum_1^{n+1} \lambda_i v_i = \sum_1^{n+1} \lambda_i (u_i + v_{n+1}) = 0$ which implies that $\sum_1^{n+1} \lambda_i u_i + \sum_1^{n+1} \lambda_i v_{n+1} = \sum_1^{n+1} \lambda_i u_i = 0$ hence $\lambda_i = 0$ for $i = 1, \dots, n$ since u_i's are linearly independent. It follows that $\lambda_{n+1} v_{n+1} = 0$ hence $\lambda_{n+1} = 0$ too.

Conversely, suppose $\sum_1^n \beta_i u_i = \sum_1^n \beta_i (v_i - v_{n+1}) = 0$ for some coefficients β_i, then we have $\sum_1^n \beta_i v_i - v_{n+1} \sum_1^n \beta_i = 0$. Let $\beta_{n+1} = -\sum_1^n \beta_i$, we have $\sum_1^{n+1} \beta_i v_i = 0$ and $\sum_1^{n+1}$ hence $\beta_i = 0$ for all i. It follows that the dimension of the affine hull is at least n. Finally, let u be some vector in the affine hull U, and let $v = u + v_{n+1}$, then $v = \sum_1^{n+1} \lambda_i v_i$ with coefficients $\sum_1^{n+1} \lambda_i = 1$ by definition. Hence $u = \sum_1^{n+1} \lambda_i v_i - v_{n+1} = \sum_1^{n+1} \lambda_i v_i - v_{n+1} \sum_1^{n+1} \lambda_i = \sum_1^{n+1} \lambda_i (v_i - v_{n+1}) = \sum_1^{n+1} \lambda_i u_i$. But $u_{n+1} = 0$ so $u = \sum_1^n u_i$. Therefore, $u_1, \dots, u_n$ spans U and the result follows. □

Theorem 2.14 (Minkowski's theorem)**.** *(Proposition 3.19 and 3.3 of [10]) Let $X \subset \mathbb{R}^n$ be a non-empty, d-dimensional closed, convex set, then for each point $a \in \partial X$, there exists a hyperplane H such that $a \in H$ and X is contained in one of the open half-spaces determined by H.*

Here a hyperplane is defined in the usual sense, an $n-1$ dimensional affine subspace of $\mathbb{R}^n$. We observe that $H \cap \operatorname{int}(X) = \emptyset$. The proof of Minkowski's theorem employs a famous theorem, namely the geometric Hahn-Banach theorem. It is one of the versions of the Hahn-Banach theorem and can be deduced from the analytic version in functional analysis. We omit the proof and see for example Gallier's book [10] for details.

Theorem 2.15 (Geometric Hahn-Banach theorem)**.** *(Theorem 3.6 of [10]) Let $X \subset \mathbb{R}^n$ be an affine space, $S \subset X$ be an non-empty, open, convex subset. Then for every affine subspace U of X with $U \cap S = \emptyset$, there exists a hyperplane H such that $U \subset H$ and $H \cap S = \emptyset$.*

Proof of theorem 2.14. If $d < n$, then X is contained in its affine hull with dimension d, hence any hyperplane contains this affine subspace would work. So suppose $d = n$, then there exists $n+1$ affinely independent points $a_0, \dots, a_n$ in X, and the convex hull of these points is contained in X. Consider the point $a = \frac{a_0 + \cdots + a_n}{n+1}$. a is clearly contained in the interior of the convex hull so X has nonempty interior. By corollary 2.11, $X = \overline{X} = \overline{\operatorname{int}(X)}$. For any point $x \in \partial X$, we have $\{x\} \cap \operatorname{int}(X) = \emptyset$. By the Hahn-Banach theorem, there exists some hyperplane H containing x and disjoint from $\operatorname{int}(X)$. In particular, $\operatorname{int}(X)$ is contained in some closed halfspace so it also contains X and this proves the theorem. □

From this proof, we also note that in general for an affine subspace $X \subset \mathbb{R}^n$, $\text{int}(X) \neq \emptyset$ if and only if the dimension of X is equal to n. Now we are ready to prove theorem 2.9.

Proof of theorem 2.9. We denote the set of extreme points of A by $E(A)$. We argue by induction on the dimension of U. For $n = 1$, a compact, convex set must be a closed interval or a singleton. The later case is trivial. In the former case, the extreme points are the end points which obviously generates the interval.

Now suppose the theorem is true for dimension $n - 1$. By theorem 2.14, for every point x in the boundary, there exists some hyperplane H containing x and contains A in one of the closed half-spaces. Since $H \cap A \subset \partial A$, each extreme point of $H \cap A$ is also an extreme point of A so we have $E(A \cap H) = E(A) \cap H$ for every such H.

By lemma 2.8, it suffices to show that $\partial A \subset \text{conv}(E(A))$. So let $x \in \partial A$ with an associated hyperplane H as mentioned above and in theorem 2.14. Since A is compact and H is closed, $H \cap A$ is also compact(of course also convex). Since H has dimension $n - 1$, by induction, we have $A \cap H = \text{conv}(E(A \cap H)) = \text{conv}(E(A)) \cap H$ which is clearly a subset of $\text{conv}(E(A))$. Therefore $a \in \text{conv}(E(A))$ as desired. □

Definition 2.16. Let U and V be vectors spaces. A map $f : U \to V$ is an **affine map** if for any affine combination $\alpha_1 u_1 + \cdots + \alpha_k u_k$ in U, $f(\alpha_1 u_1 + \cdots + \alpha_k u_k) = \alpha_1 f(u_1) + \cdots + \alpha_k f(u_k)$.

It is clear that every linear map is also affine. Another typical example of an affine map is the translation $f(x) = x + c$ for all $x \in U$ and fixed $c \in V$. The next proposition summarises the properties of affine maps

Proposition 2.17. *(Based on chapter 4 of [11]) Let $f : U \to V$ be a map between vector spaces, then the following are equivalent*

1. *f is an affine map.*
2. *$g(x) = f(x) - f(0)$ is a linear map.*
3. *There exits a linear map g and some $a \in V$ such that $f(x) = g(x) + a$.*

Proof. $1 \Rightarrow 2$ For all $x, y \in U$ and all $\alpha \in k$, $g(x+y) = f(x+y-0) - f(0) = f(x) + f(y) - f(0) - f(0) = g(x) + g(y)$ and $g(\alpha x) = f(\alpha x + (1-\alpha)0) - f(0) = \alpha f(x) + (1-\alpha) f(0) - f(0) = \alpha(f(x) - f(0)) = \alpha g(x)$.

$2 \Rightarrow 3$ Set $g = f(x) - f(0)$ and $a = f(0)$.

$3 \Rightarrow 1$ $f(\alpha_1 u_1 + \cdots + \alpha_k u_k) = g(\alpha_1 u_1 + \cdots + \alpha_k u_k) + a = \alpha_1 g(u_1) + \cdots + \alpha_k g(u_k) + (\sum_1^k \alpha_i) a = \alpha_1 f(u_1) + \cdots + \alpha_k f(u_k)$. □

By 3 of proposition 2.17, the image of any affine map is an affine set. Also, for an affine map f, every non-empty fibre $f^{-1}(y)$ is also an affine set. To see this, let $f = g + a$, let

$x_0 \in f^{-1}(v)$. Then $x \in f^{-1}(v)$ if and only if $f(x) - f(x_0) = g(x) - g(x_0) = g(x - x_0) = 0$. Thus $f^{-1}(v) = x_0+\ker(g)$ which is an affine subspace.

2.2 Convex polytopes

In this section we give an introduction of the theory of convex polytopes. The study of convex polytopes is a very ancient subject of mathematics. Although in low-dimensional cases, many results in this section are usually "intuitively" clear, the proofs are usually not obvious. Also, relaying on intuitions is not always accurate and satisfactory. Hence it is valuable and inspiring to explore the mathematical formulations rigorously.

Definition 2.18. Let $\boldsymbol{l}_1, \ldots, \boldsymbol{l}_k \in \mathbb{R}^n$, the **convex polyhedral cone** σ spanned by $\boldsymbol{l}_1, \ldots, \boldsymbol{l}_k$ is the set $\sigma = \{r_1\boldsymbol{l_1} + \cdots + r_k\boldsymbol{l}_k : r_i \geq 0, i = 1, \ldots, k\}$, or in matrix form we have $\sigma = \{V\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}^k, \boldsymbol{x}_i \geq 0, V \in \mathbb{R}^{n\times k}\}$ where the columns of V are the coordinates of $\boldsymbol{l}_1, \ldots, \boldsymbol{l}_k$. Given a finite set Y of vectors in $\mathbb{R}^n$, the cone spanned by these vectors is denoted by cone(Y).

Definition 2.19. A convex **V-polytope** is the convex hull of a finite set of points in some $\mathbb{R}^n$. A convex **V-polyhedron** is the vector sum of a V-polytope and a polyhedral cone. That is, $P = \text{conv}(V) + \text{cone}(Y) = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in \text{conv}(V), \mathbf{y} \in \text{cone}(Y)\}$ for some finite sets V and Y of points in $\mathbb{R}^n$.

Definition 2.20. A convex **H-polyhedron** is an intersection of finitely many closed half-spaces in some $\mathbb{R}^n$. A convex **H-polytope** is a bounded H-polyhedron, i.e. a convex H-polyhedron that does not contain any ray $\{x+ty : t \geq 0\}$ for any $x \in P$ and any $y \in \mathbb{R}^n$.

In other words, a convex H-polyhedron P is determined by a family of linear functionals $l_i \in (\mathbb{R}^n)^*$ and real numbers $a_i \in \mathbb{R}$ for $i = 1, \ldots, m$:

$$P = \{x \in \mathbb{R}^n : l_i(\boldsymbol{x}) \geq -a_i, i = 1, \ldots, m\}$$

Alternatively, we can put it in matrix form $P = \{\boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} \geq \mathbf{a}\}$ where A is a $m \times n$ matrix the columns of A are the matrix representations of the above linear functionals l_i. Note that a V-polytope is just a bounded V-polyhedron in the same sense as definition 2.20. Through out this project, we don't treat the empty set as a polyhedron or a polytope.

Example 2.21. Figure 2 shows that a pentagon is the convex hull of its 5 vertices, but it is also the intersection of five half-spaces corresponding to its five edges.

It turns out that these two definitions of convex polytopes are actually equivalent. One possible proof is based on an algorithm called the Fourier-Motzkin elimination which employs a large chunk of tedious linear algebra. See for example chapter 1 of Ziegler's book [8] for details. Alternatively, if we are only interested in convex polytopes rather than polyhedrons, there is a different proof via polarity(which we will give in section 2.5). Thus there is no ambiguity about what we mean by just saying "a convex polytope".

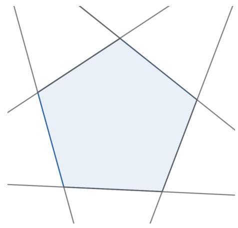


Figure 2: A pentagon

Definition 2.22. The **dimension** of a convex polytope is the dimension of its affine hull. An n-dimensional polytope is called an n-polytope. Usually we assume that an n-polytope is setting in $\mathbb{R}^n$.

Definition 2.23. A **supporting hyperplane** H of a convex polytope P is an affine hyperplane which divides $\mathbb{R}^n$ into two half-spaces such that $H \cap P \neq \emptyset$ and P is contained in one of the closed half-spaces decided by H.

Definition 2.24. A proper **face** of a convex polytope P is the intersection of P with a supporting hyperplane. The dimension of a proper face is the dimension of its affine hull.

We also regard the polytope itself as a (non-proper) face, so each face of P has dimension less or equal than n. We call 0-dimensional faces(single points) **vertices** and $(n-1)$-dimensional faces **facets**. The set of all vertices of P is denoted by $\mathrm{vert}(P)$.

Proposition 2.25 (Based on p.50-55 of [8])**.** *Let P be a convex polytope, then the following holds*

1. *P is the convex hull of its vertices, $P = conv(vert(P))$.*
2. *If P is the convex hull of a finite set V, then $vert(P) \subseteq V$.*
3. *Any face F of P is a polytope itself with $F = P\cap$ $aff(F)$ and $vert(F) = F\cap$ $vert(P)$.*
4. *The faces of F are the faces of P that contained in F.*
5. *Every intersection of faces of P is also a face of P.*

The proof of this proposition employs the Farkas lemma about the solvability of a finite system of inequalities. The proof is based on Fourier-Motzkin elimination and we omit it here. See section 1.4 of [8] for details.

Lemma 2.26 (The Farkas Lemma)**.** *[Proposition 1.8 of [8]] Let A be a $m \times d$ real matrix and $\boldsymbol{b}$ be a real, m-dimensional vector, then exactly one of the following is true:*

1. *There exists $\boldsymbol{y} \in \mathbb{R}^d$ such that $A\boldsymbol{y} = \boldsymbol{z}$.*
2. *There exists $\boldsymbol{c} \in \mathbb{R}^m$ such that $\mathbf{c}^t A \geq \mathbb{0}$ and $\mathbf{c}^t \boldsymbol{z} < 0$.*

proof of proposition 2.25. For 1 and 2, suppose $P = \mathrm{conv}(V)$. Write all points in V in a matrix whose columns are coordinates of points in V. We also denote this matrix by V. For a vector $\boldsymbol{v_i} \in V$, if it can be written as a convex combination of the other vectors in

V, then $V' = V \setminus \{\boldsymbol{v_i}\}$ also defines P. If not, then we have $\boldsymbol{v_i} \notin \text{conv}(V')$. Hence there is no (row) vector $\boldsymbol{t} \geq \boldsymbol{0}$ with $\mathbb{1}\boldsymbol{t} = 1$ such that

$$\boldsymbol{v_i} = V'\boldsymbol{t}$$

Rewrite the above, we have

$$\begin{pmatrix} \mathbb{1} \\ V' \end{pmatrix} \boldsymbol{t} = \begin{pmatrix} 1 \\ \boldsymbol{v_i} \end{pmatrix}$$

Apply the Farkas Lemma, there exist a row vector $\boldsymbol{a}$ with

$$\boldsymbol{a} \begin{pmatrix} \mathbb{1} \\ V' \end{pmatrix} \geq \mathbb{0}, \ \boldsymbol{a} \begin{pmatrix} 1 \\ \boldsymbol{v_i} \end{pmatrix} < 0$$

Hence $\boldsymbol{a}$ is in the form $(\beta, -\boldsymbol{b})$ with $\boldsymbol{b}V' \leq \beta\mathbb{1}$ and $\boldsymbol{b}\boldsymbol{v_i} > \beta$ (write out some matrices and vectors and this should be clear). It follows that for all $\boldsymbol{v_j} \in V'$, we have $\boldsymbol{b}\boldsymbol{v_j} \leq \beta$. Therefore, the hyperplane $\boldsymbol{b}\boldsymbol{x} = \boldsymbol{b}\boldsymbol{v_i}$ is a supporting hyperplane and it defines the single point $\boldsymbol{v_i}$, which must be a vertex. So we can keep removing points that are not vertices from V without changing its convex hull and we will end up with the vertex set $\text{vert}(P)$.

For 3, let F be a face of P defined by a supporting hyperplane $H = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{c}\boldsymbol{x} = c_0\}$. By the definition of an H-polytope, it is clear that F is itself a polytope. Also, clearly $F \subset P \cap \text{aff}(F)$. Every point in $P \cap \text{aff}(F)$ is also in $P \cap H$ so in F. Hence we have $F = P \cap \text{aff}(F)$.

To see $\text{vert}(F) = F \cap \text{vert}(P)$, first note a vertex v of P in F defined by a hyperplane is also a vertex of F since H is also a supporting hyperplane of F intersecting F at the single point v, so we only need to prove the opposite inclusion. Denote $\text{vert}(P)$ by V, and we prove that F is the convex hull of $F \cap V$ and the result follows from 2. Let $\boldsymbol{x} \in F$, and like we did in the proof of 1 and 2, write the points in V in matrix form. Then we have $\boldsymbol{x} = V\boldsymbol{t}$ for some column vector $\boldsymbol{t}$ with $\mathbb{1}\boldsymbol{t} = 1$. Now consider

$$\boldsymbol{c}\boldsymbol{x} = \boldsymbol{c}V\boldsymbol{t} = c_0\mathbb{1}\boldsymbol{t}$$

and we deduce that $(\boldsymbol{c}\boldsymbol{v_i} - c_0)t_i = 0$ for all $\boldsymbol{v_i} \in V$. Since $\boldsymbol{c}\boldsymbol{v_i} - c_0 \neq 0$ hence $t_i = 0$ for all vertices of P that are not in F, it follows that $\boldsymbol{x}$ is a convex combination of points in $F \cap V$, so $F = \text{conv}(F \cap V)$.

For 4, it's clear that every face $P \cap H$ of P that is contained in F is also a face of F since $F \cap H = P \cap H$. Now, suppose $F \cap H_2 = G \subseteq F$ is a face of $F = P \cap H_1$, where $H_1 = \{x \in \mathbb{R}^n : \boldsymbol{c}\boldsymbol{x} = c_0\}$ and $H_2 = \{x \in \mathbb{R}^n : \boldsymbol{b}\boldsymbol{x} = b_0\}$ are supporting hyperplanes defining F and G respectively. If $F = P$, there is nothing to prove. So let $V_1 = V \setminus \text{vert}(F)$ and we can assume $V_1 \neq \emptyset$. Since $F \subseteq H_1$, the hyperplane $H_3 = \{x \in \mathbb{R}^n : (\boldsymbol{b} + \lambda\boldsymbol{a})\boldsymbol{x} = b_0 + \lambda c_0\}$ also defines G in F for any $\lambda \in \mathbb{R}$. Choose λ large enough such that $\lambda > -\frac{b_0 - \boldsymbol{b}\boldsymbol{v}}{c_0 - \boldsymbol{c}\boldsymbol{v}}$ for all

$v \in V_1$, then all $v \in V_1$ satisfies $(\boldsymbol{b}+\lambda\boldsymbol{a})\boldsymbol{x} < b_0+\lambda c_0$. Hence H_3 is a supporting hyperplane of P too. Since $H_3 \subseteq H_1 \cap H_2$, we have $G = H_3 \cap F = H_3 \cap P$, so G is a face of P.

Finally, to prove 5, suppose F and G are two faces of P defined by $H_1 = \{x \in \mathbb{R}^n : \boldsymbol{cx} = c_0\}$ and $H_2 = \{x \in \mathbb{R}^n : \boldsymbol{bx} = b_0\}$, then the hyperplane $H_3 = \{x \in \mathbb{R}^n : (\boldsymbol{b}+\boldsymbol{a})\boldsymbol{x} = b_0 + c_0\}$ is a supporting hyperplane of P and $H_3 \cap P = F \cap G$ by construction. □

Definition 2.27. Two convex polytopes P and Q of dimension m and n are called **affinely equivalent** if there exists an affine map $f : \mathbb{R}^m \to \mathbb{R}^n$ which is bijective between P and Q. P and Q are **combinatorially equivalent** if there is a bijection between their faces such that the inclusion relation of faces are preserved.

We usually call a class of convex polytopes up to combinatorial equivalence a **combinatorial polytope**.

We would like to analyse the facets of polytopes in some detail. Given a convex H-polytope P, we assume that the defining half-spaces of P are irredundant. In other words, we cannot remove any half-space keeping the polytope unchanged. The next proposition illustrates the one-to-one correspondence between the defining half-spaces and the facets.

Proposition 2.28. *[Based on proposition 4.5 of [2]] Let $P \subset \mathbb{R}^n$ be an n-dimensional convex H-polyhedron (with nonempty interior) defined by m irredundant (closed) half-spaces*

$$P = \bigcap_{i=1}^{t} C_i$$

then these half-spaces C_i are unique up to order, and there is a one-to-one correspondence between these half-spaces and the facets F_i of P. Furthermore, each facet F_i has non-empty interior in $H_i = \partial C_i$ and $\partial P = \bigcup_{i=1}^{t} F_i$, with $F_i \subsetneq F_j$ for any $i \neq j$.

Proof. For the last statement, we have

$$\partial P = \bigcap_{i=1}^{t} C_i \backslash \mathrm{int}(\bigcap_{j=1}^{t} C_j) = \bigcap_{i=1}^{t} C_i \backslash (\bigcap_{j=1}^{t} \mathrm{int}(C_j)) = \bigcap_{i=1}^{t} C_i \cap (\bigcap_{j=1}^{t} \mathrm{int}(C_j))^c = \bigcap_{i=1}^{t} C_i \cap \bigcup_{j=1}^{t} (\mathrm{int}(C_j)^c)$$

$$= \bigcup_{j=1}^{t} (\bigcap_{i=1}^{t} C_i \cap \mathrm{int}(C_j)^c) = \bigcup_{j=1}^{t} (\bigcap_{i=1, i\neq j}^{t} C_i \cap \partial C_j) = \bigcup_{j=1}^{t} (\bigcap_{i=1}^{t} C_i \cap \partial C_j) = \bigcup_{j=1}^{t} (H_j \cap P)$$

where we used the fact that the interior of finite intersection is the intersection of the interior of components. Each $H_i = \partial C_i$ is clearly a supporting hyperplane, hence $F_i = H_i \cap C_j$ is a face.

By assumption, C_i are irredundant. So for each i, there exists some $x \in (\bigcap_{j\neq i} C_j) \setminus C_i$. Otherwise, we will have $\bigcap_{j\neq i} C_j \subset C_i$ so $P = \bigcap_{j\neq i} C_j$ which leads to a contradiction. Note that x must be in the other open half-space corresponds to C_i. Since P has non-empty interior, pick any point $p \in$int(P), then $p \in$int$(\bigcap_{j\neq i} C_j)$. By lemma 2.10, the open line segment l from x to p is contained in $\text{int}(\bigcap_{j\neq i} C_j)$. Now take $l \cap H_i$. This is non-empty since x and p are contained in two different half-spaces. It follows that $l \cap H_i \subset \text{int}(H_i \cap (\bigcap_{j\neq i} C_j)) \subset \text{int}(H_i)$. Hence each F_i has non-empty interior in $H_i = \partial C_i$, and it follows that $\dim(F_i) = \dim(H_i) = n-1$. Therefore, each F_i is actually a facet.

It remains to show that F_i's are the only facets. Suppose there exists an extra facet $F' = H \cap P$, then the supporting hyperplane H gives a closed half-space C' containing P, and C' is distinct to all existing C_i. So we can write

$$P = (\bigcap_{i=1}^{t} C_i) \cap C'$$

By the same calculation as above, we can written ∂P as

$$\partial P = (\bigcup_{i=1}^{t} F_i) \cup F' = \bigcup_{i=1}^{t} F_i$$

So $F' \subset \partial P$. We claim that F' is contained in F_i for some i. Otherwise, consider the expression $F' = (F' \cap F_1) \cup \cdots \cup (F' \cap F_n)$. However, each component $F' \cap F_i$ has dimension at most $n-2$ since they are contained in two different hyperplanes, but F' has dimension $n-1$ which is impossible. So $F' \subset F_i$ for some i. But this implies the affine hull satisfies $H' \subset H_i$ so we have $H = H_i$ which contradicts the assumption that $F' \neq F_i$. Hence the F_i's are the only facets.

By a very similar dimension argument, one shows that every convex polyhedron admits a unique irredundant decomposition, and this completes our proof. $\square$

This proposition complements proposition 2.25 and gives us a complete description of the faces of convex polytopes(polyhedrons). It is also possible to prove this result using results from linear program and its dual theorems. Briefly, we can use the complementary slackness theorem to characterise faces of P as subsystems of the defining linear system $A\boldsymbol{x} \geq \boldsymbol{a}$ and to show that every facets corresponds to a single inequality. We omit the details, see section 8.3 and 8.4 of [5] for details.

Hence every facets corresponds to a linear functional l_i, which is a vector orthogonal to the facet itself(and the associated hyperplane). Furthermore, for every facet $F_i = P \cap H_i = \bigcap_{j=1}^{t}(C_j \cap H_i)$, each $H_i \cap C_j$ gives a "half-space" if we view F_i as a polyhedron in $\mathbb{R}^{n-1}$. Hence $F_i = \bigcap_{j\neq i}^{t} D_j$ where $D_j = C_j \cap H_i$. Facets of F_i are $(n-2)$-dimensional faces of P. Repeating this process, we have a list of faces of of P in all dimensions. In particular, an $(n-k)$-dimensional face is the intersection of k hyperplanes chosen from H_i's defining P. Actually this list is complete: by the same method as in the proof of proposition 2.28, we can write $\partial P = \bigcup F_i^k$ as the union of all k-dimensional faces in our list. Any k-dimensional face F of P has affine hull an $(n-k)$-dimensional affine subspace. Also, F must be in the boundary of P. Then we have $\partial P = \partial P \cup F$. It follows that F must be in some F_i^k for some i, otherwise we deduce exactly the same contradiction by dimension except that we have finite intersection of hyperplanes rather than a single hyperplane.

It also follows that the total number of faces of a convex polytope is finite, hence the set of all faces of a convex polytope form a finite partially ordered set respect to inclusion, called the **facet poset**. The face poset forms a lattice called the face lattice.

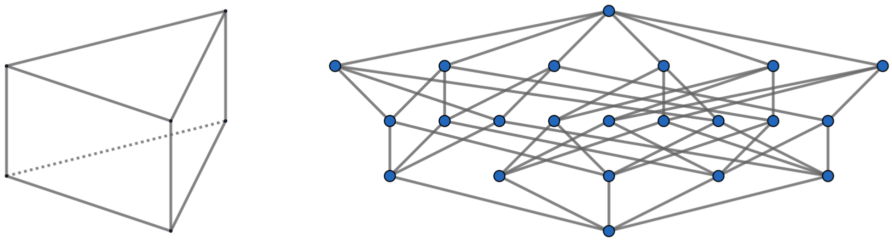


Figure 3: The face lattice of a triangular prism

Example 2.29. Figure 3 shows the face lattice of a triangular prism. The rows from top to bottom represents the empty face, 6 vertices, 9 edges, 5 faces and the non-proper face(the triangular prism itself). Every vertex is contained in three edges. There are two faces contains three edges and three faces contains four edges.

Proposition 2.30. *[Proposition 4.6 of [2]] Let $P \subset \mathbb{R}^n$ be an n-dimensional H-polyhedron (with non-empty interior), then for any $x \in \partial P$, the intersection of all faces containing x is equal to the intersection of all supporting hyperplanes of P containing x.*

Proof. Let H be a supporting hyperplane at x, then H determines a half-space C containing P with $P \cap C = P$. It follows that H is one of the H_i's in the irredundent decomposition of P or C is redundant. Suppose $A = \bigcap G_i$ is the intersection of all faces containing x, for faces G_i(at any dimension). Each G_i is determined by some hyperplanes H_i by the discussion after proposition 2.28, so we have $A = P \cap \bigcap H_i$ for some of the H_i's. Put things together, A is clearly equal to the intersection of all hyperplanes containing x. $\square$

Corollary 2.31. *[Proposition 4.6 of [2]] Let $P \subset \mathbb{R}^n$ be an n-dimensional polyhedron (with non-empty interior), then the extreme points of P are exactly the vertices.*

Proof. Suppose $a \in \partial P$ is not an extreme point, then a is contained in some line segment l in the boundary. Hence a is contained in the (relative) interior of a face of dimension at least 1. If a is a vertex, then the intersection of all supporting hyperplanes containing a should be a itself by proposition 2.30. However, in this case the intersection has at least dimension 1, hence a is not a vertex. $\square$

Example 2.32. An n-dimensional simplex Δ_n is the convex hull of $n+1$ affinely independent points in $\mathbb{R}^n$. Equivalently, Δ_n is the convex hull of $n+1$ points that do not lie on a common affine hyperplane. Any n of these points lie on a hyperplane which does not contain the remaining one point, this hyperplane H is a supporting hyperplane defining a face of Δ_n which is itself an $(n-1)$-dimensional simplex. Also, by translating and rotating H properly, we can make it into a supporting hyperplane H' intersecting with Δ_n at the remaining single point. Therefore, the vertices of Δ_n are exactly the given points. In general, any k points of the given points form an $(k-1)$-dimensional simplex.

Consider the simplex defined by the convex hull of $(1, 0, \ldots, 0)$, $\ldots$, $(0, \ldots, 0, 1)$ and $\mathbf{0}$. Here we call it the standard simplex. It's easy to verify that these points are affinely independent. In terms of formulas, we have $\Delta_n = \{(t_1, \ldots, t_n) \in \mathbb{R}^n \colon \sum_1^n t_i = 1, t_i \geq 0, \forall i\}$. In terms of an H-polytope, it is defined by the following half-spaces

$$C_i = \{\mathbf{x} \in \mathbb{R}^n \colon x_i \geq 0\}, i = 1, \ldots, n, \quad C_{i+1} = \{\mathbf{x} \in \mathbb{R}^n \colon \sum_1^n x_i \leq 1\}$$

Any two simplices with the same dimension are affinely equivalent. In particular, given another n-simplex with vertices $v_1, \ldots, v_n$, denoted by $[v_1, \ldots, v_n]$, we define an affine map f from the standard simplex to $[v_1, \ldots, v_n]$ by $f(t_1, \ldots, t_n) = \sum_1^n t_i v_i$ which is obviously an isomorphism.

2.3 Polarity

Lemma 2.33. *Let H be an affine hyperplane in $\mathbb{R}^n$ such that $0 \notin H$, then there exists a unique point h such that $H = \{a \in \mathbb{R}^n \colon h \cdot a = 1\}$.*

Proof. By elementary linear algebra, every such affine hyperplane H is the zero locus of a linear equation $a_1 x_1 + \cdots + a_n x_n = \beta$ with $\beta \neq 0$. Hence dividing by β, we get a satisfactory point $h = (\frac{a_1}{\beta}, \ldots, \frac{a_n}{\beta})$. For uniqueness, suppose h_1 and h_2 both satisfies the requirement, then $f(a) = h_1 \cdot a - 1$ and $g(a) = h_2 \cdot a - 1$ are two affine maps with kernel H. Let v be a vector in $\mathbb{R}^n \setminus H$, then $f(v) \neq 0 \neq g(v)$ and $g - \frac{g(v)}{f(v)} f$ is the 0 map. Hence $g = \frac{g(v)}{f(v)} f$. In this particular case, substitute in $a = 0$, we find out that $\frac{g(v)}{f(v)} = 1$, hence $h_1 = h_2$. □

Definition 2.34. Let $a \in \mathbb{R}^n$ be an non-zero point, the **polar hyperplane** of a, $\breve{a}$, is defined by $\breve{a} = \{b \in \mathbb{R}^n \colon a \cdot b = 1\}$. Conversely, given an affine hyperplane H in $\mathbb{R}^n$ with $0 \notin H$, the **polar point** of H, is the unique point $\breve{H} = h$, such that $H = \{a \in \mathbb{R}^n \colon h \cdot a = 1\}$.

It's clear that this construction gives a bijection between non-zero points and affine hyperplanes which does not contain 0.

Definition 2.35. Let A be a subset of $\mathbb{R}^n$, the polar set or **polar dual** of A, A^* is defined as $A^* = \{b \in \mathbb{R}^n \colon a \cdot b \leq 1 \text{ for all } a \in A\}$. A^* can also be seen as the intersection of the half-spaces $\{b \in \mathbb{R}^n \colon a \cdot b \leq 1\}$ corresponding to each polar hyperplane $\breve{a}$.

It's easily seen from the definition that the polar set A^* is closed and is always a convex set no matter the original set A is or not. Also, note that $A \subset B$ implies $B^* \subset A^*$. We are mainly interested in the polar dual of convex, closed subsets of $\mathbb{R}^n$. In this case, one verifies that $A^* = \{\partial A\}^*$.

Example 2.36. Let B_r be any open ball of some radius $r \neq 0$. Then $B_r^* = \{b \in \mathbb{R}^n : ||b|| \leq 1/r\} = B_{1/r}$.

Theorem 2.37 (Separation theorem)**.** *[corollary 3.10 of [2]]*

Let A and B be two closed, convex subsets of an affine space X with $A \cap B = \emptyset$ and A compact, then there exists a hyperplane separating A and B strictly. That is, there exists some affine hyperplane H such that $H \cap A = H \cap B = \emptyset$ and A and B are contained in the two open half-spaces defined by H respectively.

This theorem can be derived from the Hahn-Banach theorem and is a fundamental result in convex geometry. We omit the proof. One can consult Gallier's notes for details. Assuming this result, we have the following important proposition.

Proposition 2.38. *[Proposition 3.22 of [2]] Let A be a subset of $\mathbb{R}^n$, then the following holds*

1. *If A is bounded, then $0 \in int(A^*)$*
2. *If $0 \in int(A)$, then A^* is bounded.*
3. *If A is a convex, closed and $0 \in A$, then $A^{**} = A$.*

Proof. If A is bounded, then it is contained in some open ball B_r centred at 0 for some radius $r > 0$. Hence $B_r^* = B_{1/r} \subset A^*$ so $0 \in \text{int}(A^*)$. This proves 1. If $0 \in \text{int}(A)$, then $B_r \subset \text{int}(A)$ for some $r > 0$, so $A^* \subset B_{1/r}$ and this proves 2.

For 3, by definition we have $A^{**} = \{c \in \mathbb{R}^n : d \cdot c \leq 1 \forall d \in A^*\}$. First note that $A \subset A^{**}$. Suppose $b \notin A$, by the separation theorem, there exists some hyperplane H, which does not contain 0, separating $\{b\}$ and A strictly. Let h be the polar point of H, then we have $h \cdot b > 1$ and $h \cdot a < 1$ for all $a \in A$, which implies that $h \in A^*$ and hence $b \notin A^{**}$. Therefore, $A^{**} \subset A$ and this proves 3. $\square$

Proposition 2.39 (Expanding a remark in [2], p.50)**.** *Let A be a subset of $\mathbb{R}^n$, then $A^{**} = \overline{conv(A \cup \{0\})}$, the closure of its union with $\{0\}$.*

Proof. Since $A \subset A^{**}$ and $0 \in A^{**}$ we have $\overline{\text{conv}(A \cup \{0\})} \subset A^{**}$ by the closeness and convexity of A^{**}. Conversely, suppose there exists some $x \in A^{**} \setminus \overline{\text{conv}(A \cup \{0\})}$, by the separation theorem, there exists a hyperplane H separating $\{x\}$ and $\overline{\text{conv}(A \cup \{0\})}$ strictly. Let h be the polar point of H, then we have $h \cdot x > 1$ and $h \cdot y < 1$, for all $y \in \overline{\text{conv}(A \cup \{0\})}$ which implies $h \in A^*$ and hence $x \in A^{**}$, contradiction. $\square$

We would like to find out some nice properties of the polar dual of a convex polytope.

Proposition 2.40. *[Proposition 4.3 of [2]] The polar dual of of a V-polytope whose interior contains* 0 *is an H-polytope.*

Proof. Let A be a V-polytope which is the convex hull of the set of its vertices $S \subset \mathbb{R}^n$ such that $S \neq \{0\}$ and $0 \in$int(A). Suppose $S = \{a_1, \ldots, a_n\}$. By definition, we have $A^* = \{b \in \mathbb{R}^n : b \cdot \sum_1^n \lambda_i a_i \leq 1$ for $\lambda_i \geq 0$ and $\sum_1^n \lambda_i = 1\}$. But the right-hand side is equal to $\bigcap_{i=1}^n \{b \in \mathbb{R}^n : b \cdot a_i \leq 1\}$ which is an intersection of half-spaces. By Proposition 2.38, it is also bounded and the result follows. ☐

Corollary 2.41. *There is a one-to-one order-reversing correspondence between the face posets of a polytope(whose interior contains* 0*) with its polar dual polytope.*

Proof. By proposition 2.40 we see that given an n-dimensional V-polytope P, the dimension of its dual P^* is also n. Also, there is a one-to-one correspondence between the vertices of P and the facets of P^*. Hence the result is easily deduced from the discussion after proposition 2.28. ☐

Example 2.42. The polar dual of a triangular prism P is a triangular bipyramid B. It has 6 faces(all triangles), 9 edges and 5 vertices. Among those, 3 vertices are incident with 4 edges. Figure 4 partially indicates how some of the edges and faces of P turn up in B.

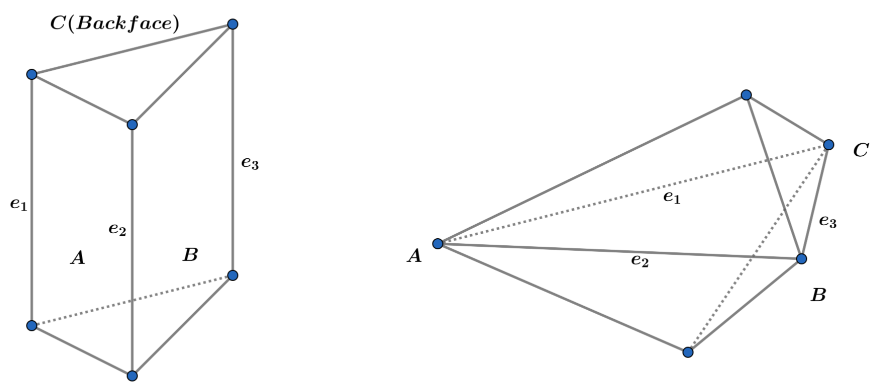


Figure 4: Triangular prism and triangular bipyramid

2.4 Simple and simplicial polytopes

In this short section we introduce two special types of polytopes: simple and simplicial polytopes.

Definition 2.43. Let P be a n-dimensional convex polytope, then P is a **simplicial** polytope if every facet of P is a simplex. P is a **simple** polytope if every vertex is contained in exactly n facets.

Proposition 2.44. *[Theorem 12.10 of [16]] If* P *and* Q *are dual polytopes of each other, then* P *is simple if and only if* Q *is simplicial.*

Proof. Suppose P is a simplicial V-polytope, then every vertex v of Q corresponds to a facet of P which is a simplex(hence contains n vertices of P). These n-vertices of P correspond to n-facets of Q containing v by corollary 2.41. Therefore Q is simple. The converse is similar. □

Of course a simplex is simplicial. Since the dual of a simplex is also a simplex, we see that simplex is also simple. The next two results give a complete classification of polytopes which are both simple and simplicial.

Lemma 2.45 (Based on section 12 of[12])**.** *All* 2*-dimensional polytopes are both simplicial and simple.*

Proof. Every facet of a 2-polytope is an edge which is a simplex, so it is simplicial. But the dual polytope of a 2-polytope is still a 2-polytope, hence the result follows from proposition 2.44. □

Theorem 2.46. *Let P be an n-dimensional polytope with $n \geq 3$, then P is both simple and simplicial if and only if P is a simplex.*

We omit the proof of this theorem due to the limitation of space. It can be proved by either employing the **vertex figure** of a polytope(i.e. cutting the polytope by a hyperplane near some verte) which requires proper definition(see [8]), or by analysing the properties of faces of simple and simplicial polytopes (see [16]).

2.5 Proof of the main theorem

Finally we get to the proof of our main theorem. After the previous preparations, our proof will be very elegant.

Theorem 2.47 (Main theorem of polytopes)**.** *[Section 4.2 of [2]] Let $P \subset \mathbb{R}^n$, then P is an H-polytope if and only if P is a V-polytope.*

Proof. Let P be an (n-dimensional)H-polytope. As mentioned before, we can assume that the $P \subset \mathbb{R}^n$ so P has non-empty interior. By the discussion after proposition 2.28 and corollary 2.31, P has a finite number of vertices which are also the extreme points of P. Since P is compact(closed and bounded) and convex, by theorem 2.9, P is the convex hull of these vertices, so a V-polytope.

Now suppose P is a V-polytope, we can assume without loss of generality that 0 is contained in P(if not, perform translations). By proposition 2.40, the polar dual P^* is an H-polytope hence a V-polytope by the first part of the proof. But P^{**} is also an H-polytope again by proposition 2.40. By proposition 2.38, $P = P^{**}$ and the result follows. □

We proved that the definitions of H-polytopes and V-polytopes are actually equivalent. In fact, this equivalence is easily generalised to H-polyhedra and V-polyhedra. In particular, proposition 2.40 can be generalised to polyhedra with a similar proof so the polar dual of a V-polyhedron is an H-polyhedron. Therefore, by exactly the same argument we did for polytopes using theorem 2.9, corollary 2.31 and proposition 2.38, we have

Theorem 2.48. *Let $P \subset \mathbb{R}^n$, then P is an H-polyhedron if and only if P is a V-polyhedron.*

Final remark: although we don't really need for the proof of 2.48, it turns out that for polyhedra, the converse of 2.40 is also true and the proof is not hard either. See section 4.3 of [2] for details.

3 Simplicial complexes

In this section we discuss a different topic: simplicial complexes. Simplicial complex is an important bridge between topology and combinatorics. The main aim of this section is to introduce a version of the Alexander duality for simplicial complexes.

3.1 Abstract simplicial complex

Definition 3.1. A **geometrical simplicial complex** P is a finite union of simplices(as discussed in example 2.32) such that

1. Every face of a simplex in P is also a simplex in P.
2. The intersection of any two simplices in P is a face of each one.

This is the "usual" simplicial complex used in the definition of simplicial homology in algebraic topology. However, we also need to introduce a set theoretical notion of an abstract simplicial complex.

Definition 3.2. Given a finite set S, an **abstract simplicial complex** K on S is a family of subsets of S such that for any $\sigma \in K$, every subset of σ is also in K. Each element of K is called an abstract simplex in K and singletons in K are called vertices. The set of all vertices is denoted by $[m]$. The simplex which contains all subsets of $[m]$ is called the full simplex on $[m]$.

We define the dimension of a simplex σ in K by $|\sigma| - 1$, and the dimension of K is the highest dimension of its simplices.

Definition 3.3. Given an abstract simplicial complex K on the vertex set $[m]$. A **geometric realization** of K is a geometric simplicial complex P such that there is a bijection between the vertex set $[m]$ and the set of all vertices of P such that every abstract simplex of K is mapped to the vertex set of some (geometric) simplex in P.

Proposition 3.4 (p.53 of [13])**.** *Every finite, d-dimensional abstract simplicial complex has a geometrical realization in $\mathbb{R}^{2d+1}$.*

Proof. Suppose K is a d-dimensional abstract simplicial complex with the vertex set $[m]$. Define an injective function $f\colon K \to \mathbb{R}^{2d+1}$ such that the image of any $2d+2$ or fewer points is a set of affinely independent vectors. This kind of f certainly exists. Hence every simplex $\sigma \in K$ is mapped to a set of affinley independent points which defines a geometric simplex σ'. Denote the union of these geometric simplices by K' and we show that K' is a geometric realization of K.

First, it is clear that if $\sigma, \beta \in K$ with $\beta \subset \sigma$, then $f(\beta) \subset f(\sigma)$. Hence every face of σ' is also in K'. Let σ_0, σ_1 be two abstract simplices in K, then the union $\sigma_0 \cup \sigma_1$ has size less or equal than $2d+9-2$. Hence $f(\sigma_0 \cup \sigma_1)$ is a set of affinely independent points by definition. So every point in $\sigma_0' \cup \sigma_1'$ has a unique convex combination representation. Therefore, a point $x \in \sigma_0' \cap \sigma_1'$ if and only if it is contained in the convex hull of $f(\sigma_0 \cap \sigma_1)$ since f is injective. Therefore, K' is a geometric simplicial complex and is a geometric realization of K by construction. □

This proposition guarantees the existence of geometric realizations of abstract simplicial complexes. However, note that in practice one sometimes need to find a geometric realization in lower dimensions rather than $\mathbb{R}^{2d+1}$.

3.2 Simplicial decomposition of convex polytopes

An intersecting connection between simplicial complexes and convex polytopes is the following:

Theorem 3.5. *[Proposition 6.8 of [2]] Every d-dimensional convex polytope P can be decomposed into a simplicial complex with out adding new vertices.*

This is called a **simplicial decomposition** of P. In other words, every convex polytope can be triangulated. Note that even if it's not immediately clear, we do need convexity. For example, every bounded subset of $\mathbb{R}^2$ whose boundary consists of finitely many polygons(homeomorphic to S^1) can be triangulated, but this fails in higher dimensions. See chapter 16 of [18] for details. To prove this theorem, we need some more definitions and a technical lemma.

Definition 3.6. Let P be a convex polytope in $\mathbb{R}^n$ and $v \in \mathbb{R}^n \setminus P$. Let F be a facet of P defined by a supporting hyperplane H. We say that F is **visible** from v if v and the interior of P are separated by the supporting hyperplane of F strictly, i.e., there exists a hyperplane H such that v and F are contained in the two open half-spaces determined by H respectively. We say that v is **beneath** F if v is contained in the open half-space which contains the interior of P determined by H.

Lemma 3.7. *[Section 5.2 of [17]] Let P and Q be two convex polytopes in $\mathbb{R}^n$ and v be a vertex of Q such that $P =$conv$(\{v\} \cup Q)$, then the following holds:*

1. *A face F of Q is a face of P if and only if there is a facet F', defined by a hyperplane H, of Q containing F such that v is beneath F.*

2. *Let F be a face of Q, then $F^* =$conv$(F \cup \{v\})$ is a face of P if and only if either $v \in$aff(F) or there exist at least one facet of Q visible from v and one facet of Q such that v is beneath it, and they both contain F.*

Proof. For 1, first note that a face of P is either also a face of Q or is the convex hull of the union of v and some face of Q. Also, a facet of Q is a facet of P if and only if v is beneath it. Hence this proves the backward direction. Conversely, suppose F is a face of P and Q, then F does not contain v. Also, there exists a facet F' of P which also does not contain v such that $F \subset F'$ and we easily see that v is beneath F'. This proves 1.

For 2, let F be a face of Q such that $F' =$conv$(\{v\} \cup F)$ is a face of P, then $F = Q \cap$aff(F'). Let x be a point in the relative interior of F and y in the interior of Q. It's clear that x, y and v are not colinear so they determines a plane E. Consider the polygon(2-dimensional polytope) $Q_0 = Q \cap E$. Then the line L determined by v and x is equal to $E \cap$aff(F'), and $F_0 = L \cap Q_0$ is either a vertex or an edge(2-dimensional face)of Q_0. If F_0 is an edge, then $v \in$aff$(F_0) \subset$aff(F). Now suppose F_0 is a vertex. By lemma 2.45, there are two edges of Q_0 containing F_0. Then it's clear that v is beneath one of them and the other one is visible from v(respect to Q_0). Choose any facets of Q containing these two edges of Q_0 respectively and this proves the forward direction of 2.

Finally, for the backward direction of 2. Suppose F is defined by a supporting hyperplane H of Q. If $v \in$aff(F), then H is also a supporting hyperplane of P with $H \cap P =$conv$(F \cup \{v\})$. Otherwise, there exist two facets F_1 and F_2 of Q with $F \subset F_1 \cap F_2$ such that F_1 is visible from v and v is beneath F_2. Let H_1 and H_2 be defining hyperplane of F_1 and F_2. By rotating H_1 and H_2 slightly about $H_1 \cap H$ and $H_2 \cap H$, we can assume that $H_1 \cap Q = H_2 \cap Q = F$. Define $H_0 =$aff$(\{v\} \cup (H_1 \cap H_2))$, we note that $H_0 \cap Q = F$ too. It follows that $H_0 \cap P =$conv$(\{v\} \cup F)$, so conv$(\{v\} \cup F)$ is a face of P. This proves the theorem. □

Proof of theorem 3.5. We proceed by induction on the number of vertices p of P. If $p = d + 1$, then P is a simplex so the theorem is true trivially. Now, for $p > d + 1$, there exists a vertex v such that the convex hull Q of the remaining vertices, which is a polytope itself, still has dimension d. By induction, Q has a simplicial decomposition D

For every facet F of Q which is visible from v, we add the simplex conv$(\{v\} \cup F)$ to the simplicial decomposition of Q to get a refined simplicial complex D'. By 1 of lemma 3.7, for facets of Q such that v is beneath F, F is also a facet of P. By 2 of lemma 3.7, facets F of Q which are visible from v are the only cases that conv$(\{v\} \cup F)$ is not a facet of P. Therefore, it follows that D' is actually a simplicial decomposition of P. □

One should note that although this theorem is intuitively clear, it is still very powerful. It reflects the face that simplicial complexes is general enough so that can be implemented to convex geometry. There are also other kinds of complexes for convex polytopes, for example, the polyhedral complexes, which are defined for the study of combinatorial manifolds. See chapter 6 of [2] for more detail.

3.3 Barycentric subdivison

Definition 3.8. Let K be an abstract simplicial complex, the **barycentric subdivision** of K is the abstract simplicial complex K' whose simplices are chains of non-empty simplices of K. That is, $\{I_1, \ldots, I_r\} \in K'$ if and only if $I_1 \subset I_2 \subset \cdots \subset I_r$ and $I_1 \neq \emptyset$. The vertices of K' are non-empty simplices of K.

One easily checks that this defines an abstract simplicial complex.

Definition 3.9. Let Δ_d be a d-dimensional geometric simplex with vertices $v_1, \ldots, v_{d+1}$, the **barycentre** of Δ_d is the point $\frac{1}{d+1}(v_1 + \cdots + v_{d+1})$.

Given an abstract simplicial complex K and a geometric realization P of K, one way of constructing a geometric realization P' of its barycentric subdivsion K' is by sending the vertices of K'(i.e. the non-empty simplices of K) to the barycentre of the corresponding simplex in P. So the simplices of P' are spanned by the barycentres of each chain $I_1 \subset \cdots \subset I_r$. It is readily checked that P' is a geometrical simplicial complex. We will see shortly that there are other ways of doing it.

Simplicial complexes and barycentric subdivisions play an important role in the early development of algebraic topology. One famous theorem is the **simplicial approximation theorem** which states that and continuous map from a finite simplicial complex K to an arbitrary simplicial complex is homotopic to a map that maps simplices to simplices linearly respect to some iterated barycentric subdivision of K. A consequence of this is the famous **Lefschetz fixed point theorem** which we don't discuss here. Chapter 2.C of [7] gives a detailed treatment.

3.4 Combinatorial Alexander duality

We first recall the original Alexander duality. We following the notations used in [7].

Theorem 3.10 (Alexander duality)**.** *[Corollary 3.45 of [7]] The material of this section is based on chapter 2.1-2.5 of [4]. Let $K \subset S^n$ be a compact, locally contractible non-empty proper subspace, then*

$$\widetilde{H}_j(S^n \setminus K) \cong \widetilde{H}^{n-1-j}(K)$$

for all j.

The proof is given in [7] and is a consequence of Poincaré duality. Here is a classical application:

Corollary 3.11. *Every compact non-orientable n-manifold M can not be embedded in $\mathbb{R}^{n+1}$.*

Proof. Suppose M can be embedded in $\mathbb{R}^{n+1}$. Consider the Alexandroff one-point compactification of $\mathbb{R}^{n+1}$ and hence view M as a subspace as a subspace of S^n. By the Alexander duality, we have

$$\widetilde{H}^i(M) \cong \widetilde{H}_{n-i-1}(S^n \setminus M)$$

But $\widetilde{H}_{n-i-1}(S^n \setminus M) = 0$ for $i \geq n$ and $\widetilde{H}_0(S^n \setminus M)$ is torsionfree since it counts the number of path-components minus 1. Therefore, $H_{n-1}(M)$ is torsion-free. However, since M is non-orientable, $H_{n-1}(M)$ must contain $\mathbb{Z}/2\mathbb{Z}$ as a subgroup. $\square$

In particular, the projective plane $\mathbb{RP}^2$ can not be embedded in $\mathbb{R}^3$ since it is not orientable.

Definition 3.12. Let K be an abstract simplicial complex on the vertex set $[m]$ and suppose K is not the full complex, then the **Alexander dual complex** of K is

$$\overline{K} = \{I \subset [m] \colon [m] \setminus I \notin K\}$$

It is easy to check that this is also an abstract simplicial complex and $\overline{\overline{K}} = K$.

Let K be an abstract simplicial complex on the vertex set $[m]$ and assume it is not the full complex. Consider the standard geometric simplex Δ_{m-1} and its boundary $\partial\Delta_{m-1}$(which is also a geometrical simplicial complex), and its barycentric subvision $\partial\Delta_{m-1}{}'$. Each face of $\partial\Delta_{m-1}{}'$ corresponds to a chain $I_1 \subset \cdots \subset I_r$ of non-empty subsets of $[m]$ with the extra restriction that $I_r \neq [m]$, and it is denoted by $\Delta_{I_1 \subset \cdots \subset I_r}$. Take the union of all faces corresponding to chains in K:

$$G(K) = \bigcup_{I_1 \subset \cdots \subset I_r, I_r \in K} \Delta_{I_1 \subset \cdots \subset I_r}$$

We observe that this is a geometric realization of K'. In particular, every chain is mapped to a face of $\partial\Delta_{m-1}{}'$.

We would like to construct a geometric realization of the dual complex $\overline{K}$. For any subset $I \subset [m]$, let $\overline{I} = [m] \setminus I$, define the following complex

$$D(K) = \bigcup_{I_1 \subset \ldots I_r, \overline{I_r} \notin K} \Delta_{\overline{I_r} \subset \cdots \subset \overline{I_1}}$$

One easily checks that this is a subcomplex of $G(K)$. Actually, it is actually a geometric realization of $\overline{K'}$. To see this, we map every vertex $\{i\}$ of $\overline{K}$ to the vertex $\overline{\{i\}} = \Delta_{\overline{\{i\}}}$ of

$\partial\Delta_{m-1}{}'$ and map every face $I \in \overline{K}$ to the vertex $\Delta_{\overline{I}}$. Hence every face $\{I_1, \ldots, I_r\}$ of $\overline{K'}$ is get mapped to the face $\Delta_{\overline{I_1}\subset\cdots\subset\overline{I_r}}$.

Lemma 3.13. *[Proposition 2.27 of [3]] $G(K)$ is a deformation retract of $\partial\Delta'_{m-1} \setminus D(K)$.*

Proof. By definition, we have $G(K) = \bigcup_{I_1\subset\cdots\subset I_r, I_r\in K} \Delta_{I_1\subset\cdots\subset I_r}$ and

$$\partial\Delta'_{m-1} \setminus D(K) = \bigcup_{I_1\subset\cdots\subset I_r, I_1\in K} \Delta_{I_1\subset\cdots\subset I_r}$$

and we see that the essential difference is in the second expression we only require $I_1 \in K$. Therefore, for $\Delta_{I_1\subset\cdots\subset I_r}$ with $I_r \notin K$, we can just delete an element from I_r repeatedly until it is in K. Geometrically, this corresponds to the projection of a face to its "subfaces" in $\partial\Delta'_{m-1}$ and this process is a (strong) deformation retract since the subsets I that are already in K remains unchanged. Hence the result follows. □

Finally, we are ready to state the combinatorial version of the Alexander duality.

Theorem 3.14. *[Combinatorial Alexander Duality][Corollary 2.28 of [3]] Given any abstract simplicial complex K on $[m]$ which is not full, then there is an isomorphism between the simplicial (reduced) homology and cohomology groups(given some coefficient ring)*

$$\widetilde{H}^j(K) \cong \widetilde{H}_{m-3-j}(\overline{K})$$

for all $-1 \leq j \leq m-2$.

Proof. $\partial\Delta'_{m-1}$ is homeomorphic to the sphere S^{m-2} since $\partial\Delta_{m-1}$ is homeomorphic to S^{2-m}. By lemma 3.13, we have

$$\widetilde{H}^j(K) = \widetilde{H}^j(G(K)) = \widetilde{H}^j(\partial\Delta'_{m-1} \setminus D(K))$$

Note that $\partial\Delta'_{m-1}$ is homeomorphic to the sphere S^{m-2} since $\partial\Delta_{m-1}$ is homeomorphic to S^{2-m}. Therefore

$$\widetilde{H}^j(\partial\Delta'_{m-1} \setminus D(K)) = \widetilde{H}^j(S^{m-2} \setminus D(K))$$

Now apply the usual Alexander duality 3.10, we get

$$\widetilde{H}^j(S^{m-2} \setminus D(K)) \cong \widetilde{H}_{m-3-j}(\overline{K})$$

□

This version of the Alexander duality completely relies on combinatorial data. Although it is as significant as the original version in terms of topology, it is still intersecting as a bridge between combinatorics and topology. The following example illustrates the concept of Alexander dual and how it fit with topology together.

Example 3.15. Consider the torus T^2 which admits the following simplicial complex structure K on the set $[m] = \{A, B, C, D, E, F, G\}$ so we have $m = 7$. It is tedious but ele-

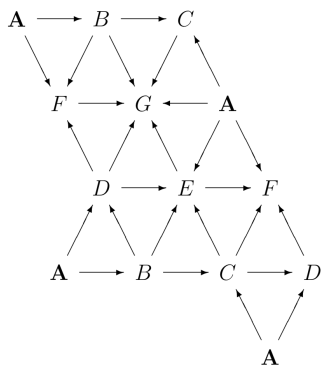


Figure 5: A Triangulation of T^2

mentary to calculate the simplicial cohomology groups of T^2 using the above triangulation. But by cellular cohomology we easily get $\widetilde{H}^2(K) = \mathbb{Z}$, $\widetilde{H}^1(K) = \mathbb{Z}^2$ and 0 otherwise.

By direct, long calculation(which we omit here), we can list the Alexander dual $\overline{K}$ of K. In particular, $\overline{K}$ has 21 simplices of size 4, 3 simplices of size 5 and 1 simplices of size 7. By theorem 3.14, we have $\widetilde{H}_{4-j}(\overline{K}) = \widetilde{H}^j(K)$. So

$$\widetilde{H}_n(\overline{K}) = \begin{cases} \mathbb{Z} & n = 2 \\ \mathbb{Z}^2 & n = 3 \\ 0 & \text{otherwise} \end{cases}$$

In particular, one can prove that $\overline{K}$ has a geometric realization: the wedge sum of S^2 and the Moore space of $\mathbb{Z}^2$ with $n = 3$, by using the Hurewicz theorem.

4 Toric varieties as topological spaces

Toric varieties are important objects in algebraic geometry which captures geometric and algebraic information via combinatorial data. In this section, we treat toric varieties as topological spaces and focus on their connections with convex polytopes and topology. As we will discuss, for every convex polytope P in $\mathbb{R}^n$, there is an associated toric variety determined by the normal vectors corresponding to its facets. It turns out that this toric variety has an easy topological description. We will study the topology of toric varieties by calculating its homology groups via hands-on examples.

4.1 Classical definitions and constructions

Before tackling topology, we introduce the algebraic definition of toric varieties for completeness. For an introductory discussion of this subject, refer to [1]. We also follows the language and conventions of algebraic geometry as in [6].

Definition 4.1. The **algebraic torus** $(\mathbb{C}^*)^n$ is the direct product of n multiplicative groups $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ of complex numbers. The **topological torus** T^n is a subgroup of $(\mathbb{C}^*)^n$, $T^n = \{(e^{2\pi i \phi_1}, \dots, e^{2\pi i \phi_n}) \in \mathbb{C}^n : \phi_i \in \mathbb{R}\}$

Note that the algebraic torus $(\mathbb{C}^*)^n$ is isomorphic to $\mathbb{C}^n \setminus V(x_1 x_2 \dots x_n)$, where $V(x_1 x_2 \dots x_n)$ denoting the vanishing set of the polynomial $x_1 x_2 \dots x_n$. Also, the algebraic torus acts naturally on itself by complex multiplication. This leads to the definition of a toric variety.

Definition 4.2. A **toric variety** is an algebraic variety M such that $(\mathbb{C}^*)^n$ is Zariski open in M and the natural action of $(\mathbb{C}^*)^n$ on itself extends to an action on M, which is given as a morphism of varieties.

Example 4.3. The simplest example is $\mathbb{C}^n$, which contains $(\mathbb{C}^*)^n$ as a Zariski open subset. Since the coordinate ring of $(\mathbb{C}^*)^n$ is $\mathbb{C}[x_1, \dots, x_n]$, the extended action is obviously a morphism.

Example 4.4. Consider the projective space $\mathbb{P}^n$ with homogeneous coordinates $x_0, \dots, x_n$. The map from $(\mathbb{C}^*)^n$ to $\mathbb{P}^n$ defined by $(t_1, \dots, t_n) \to (1, t_1, \dots, t_n)$ identifies $(\mathbb{C}^*)^n$ with the Zariski open set $\mathbb{P}^n \setminus V(x_1 x_2 \dots x_n)$. Hence the action

$$(t_1, \dots, t_n) \bullet (a_0, a_1, \dots, a_n) = (a_0, t_1 a_1, \dots, t_n a_n)$$

which proves that $\mathbb{P}^n$ is a toric variety.

Example 4.5. Another example is the vanishing set of $x^2 - y$, $V(x^2 - y) \subset \mathbb{C}^2$. $(\mathbb{C}^*)^n$ is isomorphic to the subvariety $V_{xy} = \{(t, t^2 : t \neq 0)\}$ via the map $t \to (t, t^2)$. V_{xy} is an open subset of $V(x^2 - y)$. Furthermore, the action extends to $V(x^2 - y)$ by $(t, t^2) \bullet (a, b) = (ta, t^2 b)$ which is obviously a morphism.

We would like to draw the connections between toric varieties and convex polytopes. To do this we introduce fan language. Recall the definition of a convex polyhedral cone discussed in definition 2.18. A cone is itself a V-polyhedron so we can actually define the faces of a cone as we did for polytopes. By theorem 2.48 and proposition 2.44, every face of a cone is a cone itself.

Definition 4.6. A **fan** Σ is a set of polyhedral cones in $\mathbb{R}^n$ such

1. Every face of a cone in Σ is also a cone in Σ.
2. The intersection of two cones in Σ is a face of each one.

It is easy to check that a fan forms a partially ordered set. To be able to talk about toric varieties, we need the notion of a dual cone. Recall that in general given a lattice N in $\mathbb{R}^n$, the dual lattice $M = Hom(N, \mathbb{Z}) \cong \mathbb{Z}^n$. We will only work with the lattice $\mathbb{Z}^n$ whose dual is itself.

Definition 4.7. Let σ be a polyhedral cone, the **dual cone** $\check{\sigma}$ is defined by

$$\check{\sigma} = \{u \in (\mathbb{R}^n)^* \colon \langle u, v\rangle \geq 0, \forall v \in \sigma\}$$

Suppose given a cone $\sigma = \{V\boldsymbol{x} \colon \boldsymbol{x} \in \mathbb{R}^k, \boldsymbol{a}_i \geq 0, V \in \mathbb{R}^{n\times k}\}$, then the dual cone $\check{\sigma} = \{\boldsymbol{y} \in \mathbb{R}^n \colon \boldsymbol{y}^t V\boldsymbol{x} \geq 0\} = \{\boldsymbol{y} \in \mathbb{R}^n \colon \boldsymbol{x}^t V^t \boldsymbol{y} \geq 0\} = \{\boldsymbol{y} \in \mathbb{R}^n \colon V^t\boldsymbol{y} \geq 0\}$ is an H-polyhedral. Hence the dual cone of a cone is also a cone. Note that the notation of dual cone is actually a special case of definition 2.35.

To define the associated toric variety of the fan Σ, we need the following simple lemma from convex geometry.

Lemma 4.8. *(Gordan's lemma)(p12, proposition in [1]) If σ is a polyhedral cone spanned by rational points, then $S_\sigma = \check{\sigma} \cap M$ is a finitely generated semigroup.*

Proof. Since σ is rational, the dual cone $\check{\sigma}$ is also rational with some integer generators $u_1, \dots, u_s$ in S_σ. Let $K = \{\sum_{i=1}^k t_i u_i \colon t_i \in [0,1]\}$. This is a compact subset, hence the intersection $K \cap M$ is finite. Let $u \in S_\sigma$ with $u = \sum_{i=1}^k r_i u_i$, we can write each r_i as the sum of two non-negative numbers a_i and b_i such that $a_i \in [0,1]$. Hence we have $u = \sum_{i=1}^k a_i u_i + \sum_{i=1}^k b_i u_i = \sum_{i=0}^k b_i u_i$ where we set $b_0 u_0 = \sum_{i=1}^k a_i u_i$. Hence we have written u as a sum of vectors in $K \cap M$, so $K \cap M$ generates S_σ. $\square$

By Gordan's lemma, S_σ determines a group ring

$$A = \mathbb{C}[S_\sigma] = \{f \colon S_\sigma \to \mathbb{C} \colon \text{all but finitely many elements of } S_\sigma \text{ get mapped to } 0\}$$

which is a finitely generated commutative $\mathbb{C}$-algebra with basis $\{\chi_u \colon u \in S_\sigma\}$ and multiplication $\chi_u \cdot \chi_v = \chi_{u+v}$ where $\chi_u(g) = 1$ if $g = u$ and 0 otherwise.

Every finitely generated commutative $\mathbb{C}$-algebra is isomorphic to some quotient of a polynomial ring $\mathbb{C}[x_1, \dots, x_r]/I$. Therefore, A defines an affine variety over $\mathbb{C}$ as the spectrum V_σ=Spec(A) with coordinate ring A. Similarly, for every face τ of σ, the dual face $\check{\tau}$ is contained in $\check{\sigma}$, hence defines an affine variety with a morphism $V_\tau \to V_\sigma$ induced by the inclusion of corresponding group rings. It can be check that this morphism is an inclusion of a Zariski open set (see page 15-23 of [1]). Hence one can define a variety on the fan Σ, denoted by M_Σ, by gluing(identifying) all varieties corresponds to cones of Σ together. It is also explained in the first chapter of [1] that how to check M_σ is actually a toric variety(which we omit here). Furthermore, there is a one-to-one correspondence between toric varieties of complex dimension n and fans in $\mathbb{R}^n$.(see section 5.1 of [4] and [1]). Thus the associated fan of a toric variety actually captures all information of the variety. Therefore, we call M_Σ the toric variety corresponding to the fan Σ.

Example 4.9. Consider the standard basis $e_1 = (1, 0)$ and $e_2 = (0, 1)$ of $\mathbb{R}^2$ and the fan Σ in $\mathbb{R}^2$ containing the following three cones: σ_0 generated by e_1 and e_2, σ_1 generated by e_2 and $-e_1 - e_2$ and σ_2 generated by e_1 and $-e_1 - e_2$. We find the toric variety of this fan Σ. As mentioned before, we choose lattice $\mathbb{Z}^2$ in $\mathbb{R}^2$ so $M = \mathbb{Z}^2$. It is easy to see that the dual cone $\check{\sigma_0}$ is isomorphic to σ_0. $A_0 = \mathbb{C}[S_{\sigma_0}]$ is generated by χ_{e_1} and χ_{e_2}. Therefore A_0 is isomorphic to $\mathbb{C}[x, y]$ by sending χ_{e_1} to x and χ_{e_2} to y via the homomorphism

$$\phi \colon A_0 \to \mathbb{C}[x, x^{-1}, y, y^{-1}]$$

Therefore, the variety V_0 of σ_0 is just $\mathbb{C}^2$ with coordinates x, y. Similarly, $\check{\sigma_1}$ is generated by $-e_1$ and $(-1, 1) = -e_1 + e_2$. Hence via the same homomorphism we have $\phi(\chi_{-e_1}) = x^{-1}$ and $\phi(\chi_{-e_1+e_2}) = x^{-1}y$. So A_1 is isomorphic to $\mathbb{C}[x^{-1}, x^{-1}y]$ and the variety of σ_1 is isomorphic to $\mathbb{C}^2$ too. Finally, consider σ_2 with dual cone $\check{\sigma_2}$ generated by $-e_2$ and $e_1 - e_2$. By a similar calculation, one obtain that A_2 is isomorphic to $\mathbb{C}[y^{-1}, xy^{-1}]$ and V_3 is also isomorphic to $\mathbb{C}^2$.

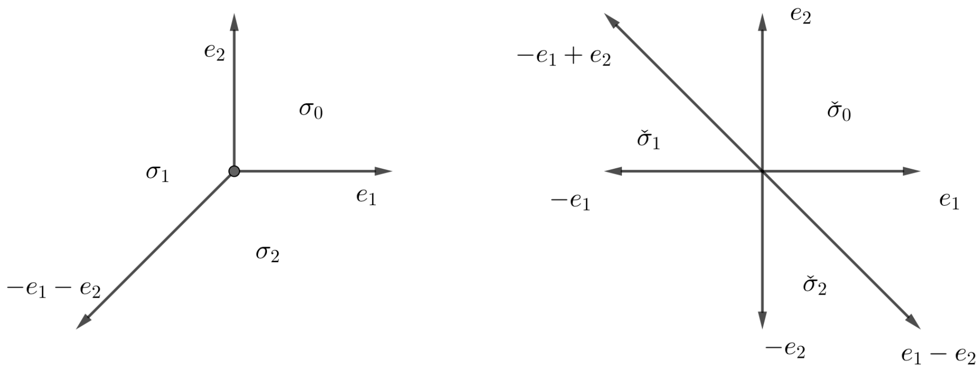


Figure 6: The fan Σ

So we have three copies of affine open sets $\mathbb{C}^2$: V_0, V_1 and V_2. Consider the complex projective space $\mathbb{CP}^2$ with coordinates (z_0, z_1, z_2) as in example 4.4. By setting $x = \frac{z_1}{z_0}$ and $y = \frac{z_2}{z_0}$, we see that V_0, V_1 and V_2 correspond to the affine charts $\{z_i \neq 0\}$ of $\mathbb{CP}^2$ for $i = 0, 1, 2$ respectively. Therefore, they glued together naturally and lead to $M_\Sigma = \mathbb{CP}^2$. We will come back to this example in section 4.3 later from a topological point of view.

4.2 Topological construction

Given an n-dimensional H-polytope P defined by a set of vectors $l_1, \ldots, l_m$ such that the vertices of P are in the lattice $\mathbb{Z}^n$. As discussed before, the vector l_i is orthogonal to the correspondence hyperplane whcih determines a half-space. Hence each l_i is orthogonal to a facet which contains at least n vertices. It follows that we can choose the coordinates of l_i to be integral and hence also primitive by scaling.

Define a complete fan $\Sigma(P)$ whose cones are generated by the normal vectors $l_{i_1}, \ldots, l_{i_k}$ such that the corresponding facets $F_{i_1} \ldots, F_{i_k}$ have non-empty intersection in P for $k = 1, \ldots, m$. Denote the corresponding toric variety constructed from $\Sigma(P)$ by $M_P = M_{\Sigma(P)}$.

We are interested in the topological information of this toric variety. We follow construction 5.5 in [3]. Suppose we have a simple polytope P. For each point p in the polytope P, denote $G(p)$ as the smallest face containing p in its interior. The normal space $N(p)$ of $G(p)$ is spanned by the codim$G(p)$ vectors l_i orthogonal to $G(p)$. If we identify the torus T^n with $\mathbb{R}^n/\mathbb{Z}^n$, then $N(p)$ projects to a subtorus $T(p)$ of T^n. It turns that(details in section 4.1 of [1]) the toric variety constructed from P, as a topological space, can be viewed as

$$M_P = T^n \times P/\sim$$

where $(t_1, p_1) \sim (t_2, p_2)$ if $t_1 = \lambda t_2$ and $p_1 = p_2$ for some $\lambda \in T(p)$. No proper face contains the any interior points, so for interior points the smallest face would be the whole polytope. Also, for a vertex point contained in n facets, all normal vectors l_i spans the whole space $\mathbb{R}^n$(since $\Sigma(P)$ is a complete fan), hence projects to the original torus T^n.

In the remaining of this chapter, we will stick to this topological description.

4.3 A worked example

We consider a specific example of the construction in 4.2. We will se that even if working with very simple polygons in $\mathbb{R}^2$ is already interesting.

Example 4.10. Consider a right triangle P, viewing as a 2-dimensional simple polytope, in $\mathbb{R}^2$ as figure 7 shows.

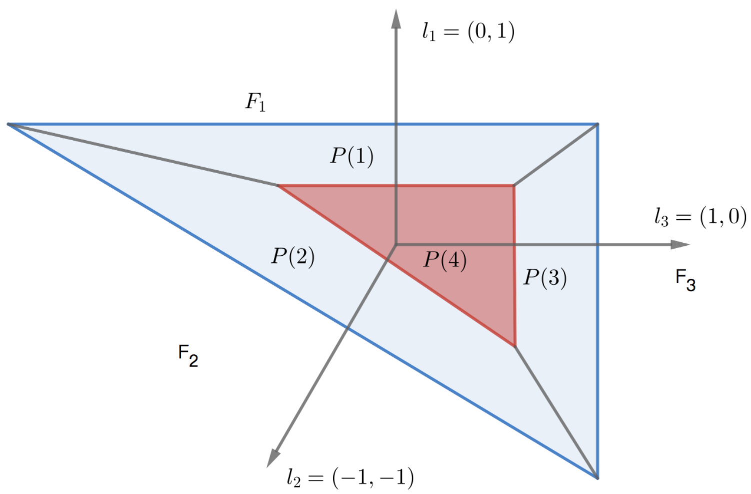


Figure 7: M_P as a topological space

Denote the three edges(facets) of P by F_1, F_2 and F_3, and the corresponding normal vectors by $l_1 = (0,1)$, $l_3 = (1,0)$ and $l_2 = (-1,-1)$ respectively. By our construction in section 4.2, the subspace spanned by l_1 projects to the subtorus $T(1) = 1 \times S^1$. The subspace spanned by l_2 projects to the subtorus $T(2) = \Delta T^2$, the diagonal of $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. Finally, the subspace spanned by l_3 projects to the subtorus $T(3) = S^1 \times 1$. So the corresponding toric variety M_P is defined as

$$M_P = T^2 \times P^2/\sim$$

where for any $\lambda \in \mathbb{R}$, $((s,t),x) \sim ((s,\lambda t),x)$ for points in F_1, $((s,t),x) \sim ((\lambda s,\lambda t),x)$ for points in F_2 and $((s,t),x) \sim ((\lambda s,t),x)$ for points in F_3. At vertices, one checks that $((s,t),x)$ are identified for all (s,t). For example, at the upper right corner, we have $((s,t),x) \sim ((s,1),x) \sim ((1,1),x)$ and similar for the other two vertices.

To understand M_P, we calculate its homology groups. As the figure shows, we divide the triangle into four parts $P(1), P(2), P(3)$ and $P(4)$. $P(4)$ is contained in the interior of P while the other three contain a facet respectively. Denote $P'(4) = P(1) \cup P(2) \cup P(3)$, which is homotopy equivalent to the boundary $\partial P \cong S^1$. We have $P = P(4) \cup P'(4)$ hence $M_P = M_{P(4)} \cup M_{P'(4)}$.

Note that $M_{P(4)}$ is not guaranteed to have a toric variety structure, but for our calculation we only need it to be a topological space. We stick to the notation "$M_{P(4)}$" for convenience.

There is no relations on $P(4)$ so $M_{P(4)} = P(4) \times T^2$ which is homotopy equivalent to T^2 since $P(4)$ is contractible. The task is to find $M_{P'(4)}$. Clearly, $M_{P'(4)}$ is homotopy equivalent to $M_{\partial P}$ via deformation retracting $P'(4)$ to ∂P. We consider the three edges separately. For the upper edge F_1, M_{F_1} is given by a cylinder with upper and lower section are collapsed to a point respectively, hence is homeomorphic to S^2. Similarly, we have $M_{F_2} \cong M_{F_3} \cong S^2$. They are glued together by attaching the "vertices" together like a triangle as shown in the below figure 10 (a). To analyse this space, we need some topological lemmas:

Lemma 4.11. *Let $G = G(V,E)$ be a finite graph with vertex set V and edge set E, then G is homotopy equivalent to the wedge sum of n cycles, $\bigvee_{i=1}^{n} S^1$ where $n = |E| - |V| + 1$.*

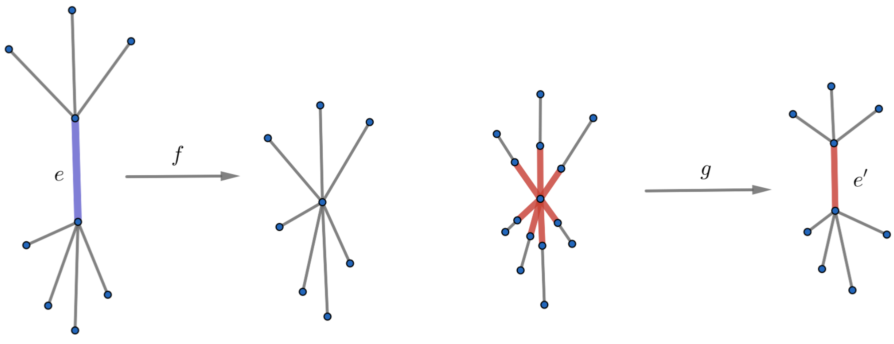
Proof. We first focus on a local case. Suppose we would like to collapse an edge of G and denote the resulting space by G/e and we claim that this does not affect the homotopy type. Define f be the collapsing map of edge e as the figure 8 shows, and defined g to be the following map: collapsing the red segments in each side together to give a new edge e' as figure 10 (c) and (d) shows.

We construct a homotopy equivalence as following:

1. Define $H\colon G/e \times [0,1] \to G$ by collapsing a proportion of the read segments in (c) gradually and then stretch the resulting new edge appropriately. If we arrange this

operation carefully, we can make it such that $H(x,0) = Id_G$ and $H(x,1) = f \circ g$.

2. Define $F\colon G \times [0,1] \to G/e$ by collapsing a proportion of the red edges and contracting a proportion of the middle edge e(from upper half and lower half separately and proportionally) as indicated in Figure 9 by dotted lines. Again, we can choose the maps such that F is a homotopy between identity and $g \circ f$.



(a) (b) (c) (d)

Figure 8: Maps f and g

Therefore, we see that collapsing edges does not affect homotopy type. Now the result follows easily: every edge is contractible so we can assume that our graph is made up of cycles that does not contain any subcircles by contracting edges not in a circle. This does not change the difference $|E| - |V|$. Furthermore, we can also assume that every two such circles share an edge since if they just share a vertex, we can just extend this vertex to an edge(and hence adding one edge and one vertex) without changing the homotopy type. Now suppose there are n such circles in total, then there are $n-1$ sharing edges in total. Hence it's easy to see that $n = |E| - |V| + 1$. Collapsing the sharing edges to one vertex, then each circle gives a copy of S^1 and the result follows. $\square$

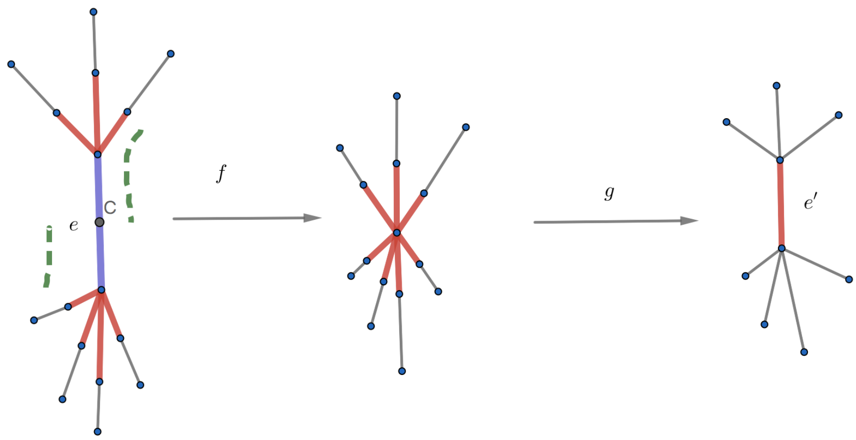


Figure 9: The homotopy F

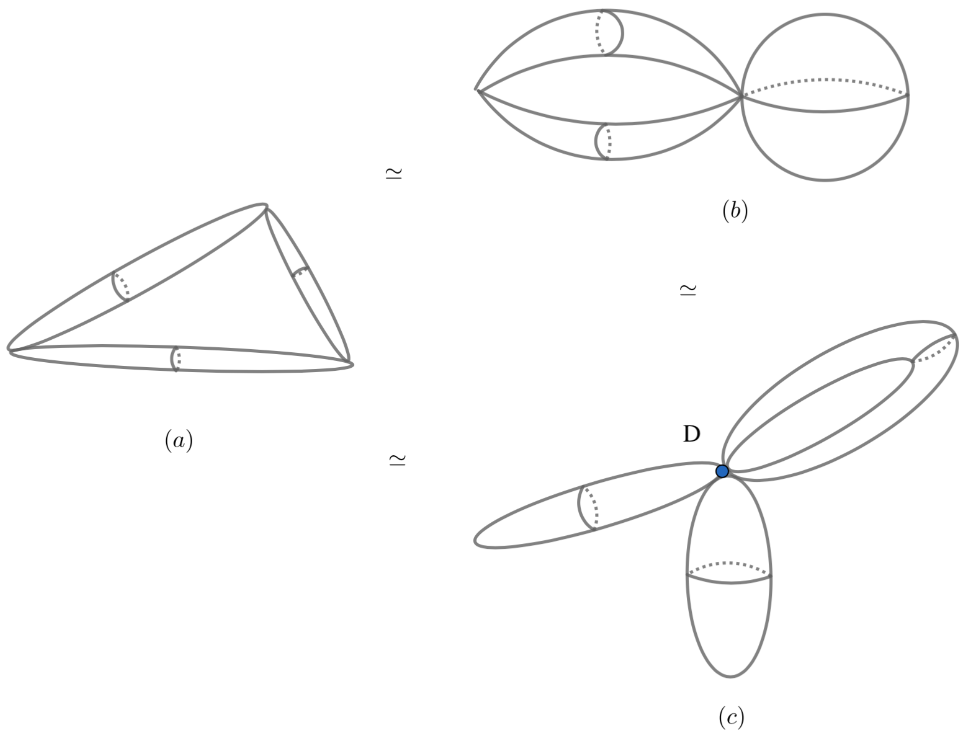


Figure 10: Homotopy equivalences in lemma 4.12

We would apply this lemma to our case where we replace the edges by copies of S^2(sausages).

Lemma 4.12. *Suppose SG is a topological space obtaining from a finite graph by replacing edges with copies of S^2 and vertices are replaced by the north and south poles of S^2, then SG is homotopy equivalent to $\bigvee_{i=1}^{n} S^2 \vee \bigvee_{j=1}^{m} S^1$ where m is the number of spheres S^2 and $n = |E| - |V| + 1$ as in lemma 4.11.*

Proof. First choose a longitude in each S^2 joining the north and south pole, then SG is reduced to the original graph G. Again, we look the local case as in the example we are interested in as figure 10 (a) shows. Similar to lemma 4.11, if we collapse alongside a longitude of one of the spheres, we get a space like figure 10 (b). If we further collapse alongside the chosen longitude of one more sphere, we get a wedge of two spheres and an "air balloon" as figure 10 (c). We call this balloon B, and we claim that B is homotopy equivalent to a wedge of S^1 and S^2.

To see this, we first extend the point D to an edge to get B' as figure 11 shows. This is clearly a homotopy equivalence. Next, we collapse along the red longitude to get B''. Conversely, if we start from B'' by collapsing the green and blue parts, and the rest part of S^1 gives back the edge e in B'. So we get B' back. Similar to the argument in lemma 4.11, the composition of these two maps are homotopy to the identities so we have $B' \simeq B''$.

In general case, we see that collapsing longitudes gives us n copies of S^1 as in lemma 4.11, but with m extra copies S^2, wedged together. $\square$

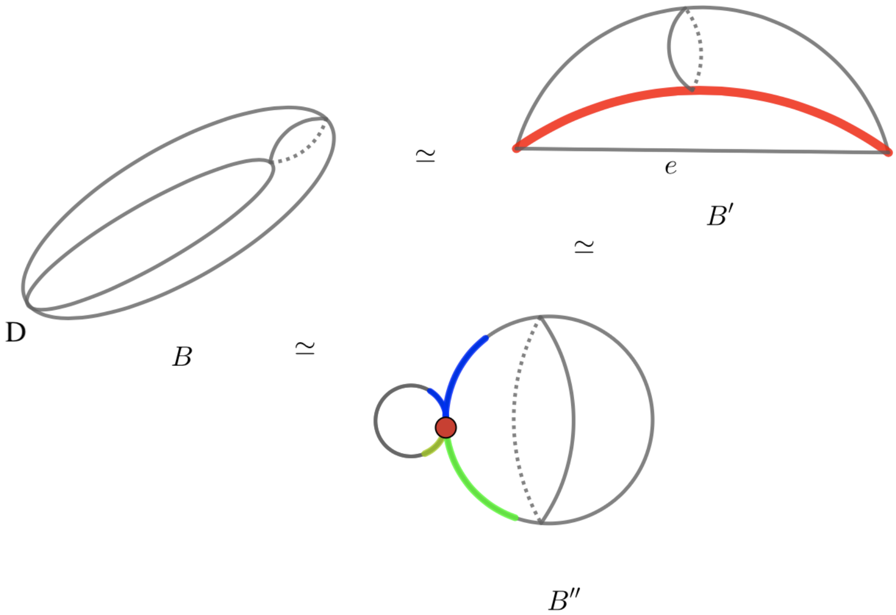


Figure 11: Homotopy equivalences in lemma 4.12

Now back to our example, by lemma 4.12, $M_{P(4)'}$ is homotopy equivalent to $S^1 \vee S^2 \vee S^2 \vee S^2$. We want to use the Mayer-vietoris sequences, so consider $M_\cap = M_{P(4)} \cap M_{P'(4)}$ which is easily seen to be homeomorphic to $S^1 \times T^2$ since no relation is imposed in the interior of P. By the results from the appendix:

$$H_n(M_{P(4)} = T^2) = \begin{cases} 0 & n \geq 3 \\ \mathbb{Z} & n = 0, 2 \\ \mathbb{Z}^2 & n = 1 \end{cases} \text{ and } H_n(M_{P(4)'} = M_{\partial P} = S^1 \vee S^2 \vee S^2 \vee S^2) = \begin{cases} 0 & n \geq 3 \\ \mathbb{Z} & n = 0, 1 \\ \mathbb{Z}^3 & n = 2 \end{cases}$$

To find out the homology groups of $S^1 \times T^2$, we apply the Kunneth formula. In particular, we have

$$H_2(M_\cap) = (H_0(S^1) \otimes H_2(T^2)) \oplus (H_1(S^1) \otimes H_1(T^2))$$

$$H_1(M_\cap) = (H_0(S^1) \otimes H_1(T^2)) \oplus (H_1(S^1) \otimes H_0(T^2))$$

$$H_n(M_\cap = S^1 \times T^2) = \begin{cases} \mathbb{Z} \otimes \mathbb{Z} = \mathbb{Z} & n = 3 \\ (\mathbb{Z} \otimes \mathbb{Z}) \oplus (\mathbb{Z} \otimes \mathbb{Z}^2) = \mathbb{Z}^3 & n = 2 \\ (\mathbb{Z} \otimes \mathbb{Z}^2) \oplus (\mathbb{Z} \otimes \mathbb{Z}) = \mathbb{Z}^3 & n = 1 \\ \mathbb{Z} \otimes \mathbb{Z} = \mathbb{Z} & n = 0 \end{cases}$$

Hence we have all inputs of the Mayer-vietoris sequences and we have a long exact sequence:

$$\begin{tikzcd}
0 \arrow[r] & 0 \arrow[r] & H_4(M_P) \arrow[dll] \\
\mathbb{Z} \arrow[r] & 0 \oplus 0 \arrow[r] & H_3(M_P) \arrow[dll] \\
\mathbb{Z}^3 \arrow[r, "\alpha"] & \overset{\sigma_3}{\mathbb{Z} \oplus \mathbb{Z}^3} \arrow[r, "\phi"] & H_2(M_P) \arrow[dll] \\
\mathbb{Z}^3 \arrow[r, "\beta"] & \overset{\sigma_2}{\mathbb{Z}^2 \oplus \mathbb{Z}} \arrow[r, "\phi'"] & H_1(M_P) \arrow[dll] \\
\mathbb{Z} \arrow[r, "\gamma"] & \overset{\sigma_1}{\mathbb{Z} \oplus \mathbb{Z}} \arrow[r, "\phi''"] & H_0(M_P) \arrow[r] & 0
\end{tikzcd}$$

From the long exact sequence, It is immediate that $H_4(M_P) = \mathbb{Z}$. To calculate the remaining groups, we need to understand the maps α, β and γ explicitly. In the first factor we have

$$H_n(S^1 \times T^2) = (H_0(S^1) \otimes H_n(T^2)) \to H_0(P(4)) \otimes H_n(T^2)$$

$$(H_1(S^1) \otimes H_{n-1}(T^2)) \to H_1(P(4)) \otimes H_{n-1}(T^2) = 0$$

for $n = 0, 1, 2$. The first map is just an isomorphism induced by inclusion since S^1 is contained in $P(4)$ which is contractible and the second map is obviously 0.

For the second factor, we need to treat three cases separately. For $n = 0$, we have

$$H_0(S^1) \otimes H_0(T^2) = \mathbb{Z} \to H_0(M_{\partial P}) = \mathbb{Z}$$

which is obviously the zero map since T^2 are quotient out in three sides of the boundary. Hence γ is the inclusion from $\mathbb{Z}$ to $\mathbb{Z}^2$.

For $n = 1$, we have

$$(H_0(S^1) \otimes H_1(T^2)) \oplus (H_1(S^1) \otimes H_0(T^2)) = \mathbb{Z}^2 \oplus \mathbb{Z} \to H_1(M_{\partial P}) = H_1(S^1) = \mathbb{Z}$$

where $\mathbb{Z}^2$ has generators $h_1 = 1 \otimes (1, 0)$ and $h_2 = 1 \otimes (0, 1)$, and $\mathbb{Z}$ has a generator $h = 1 \otimes 1$. T^2 is mapped to a copy of S^1 in S^2 in $M_{\partial P}$ which is contractible to a point in S^2. Hence there is no 1-homology on S^2, so h_1 and h_2 are sent to 0. S^1 is mapped to a longitude which is homotopy equivalent to S^1 in $M_{\partial P}$, hence the generator of $H_1(S^1) \otimes H_0(T^2)$ is just mapped by the inclusion, so h is sent to 1. It follows that β is actually an isomorphism.

Finally, for $n = 2$, the analysis of α requires more work. We have

$$(H_0(S^1) \times H_2(T^2)) \oplus (H_1(S^1) \otimes H_1(T^2)) \to H_2(M_{\partial P}) = H_2(S^2) \oplus H_2(S^2) \oplus H_2(S^2)$$

where $\mathbb{Z}$ has a generator $g = 1 \otimes 1$ and $\mathbb{Z}^2$ has two generators $g_1 = 1 \otimes (1,0)$ and $g_2 = 1 \otimes (0,1)$. Also, each copy of S^2 corresponds to an edge of P. For the second coordinate, g is sent to $(g,(1,1,1))$ since g corresponds to a copy of generator in each of the three spheres S^2 in $M_{\partial P}$. To find out the image of g_1 and g_2, recall that for each pieces $P(1), P(2)$ and $P(3)$, we have the projection map for $i = 1,2,3$:

$$T^2 \to T^2/T(i) \cong S^1$$

which induces a map of homology

$$H_1(T^2) = \mathbb{Z} \oplus \mathbb{Z} \to H_1(T^2/T(i)) = \mathbb{Z}$$

with kernel generated by $(0,1)$, $(1,1)$ and $(1,0)$ for $i = 1,2$ and 3 respectively. Therefore, for $i = 1$, the generator $g_1 = 1 \otimes (1,0)$ in $M_\cap$ corresponds to 1 in M_{F_1}, 1 in M_{F_2} and 0 in M_{F_3}. Similarly, g_2 corresponds to 0 in M_{F_1}, -1 in M_{F_2} and 1 in M_{F_3}. Hence we have

$$\begin{aligned}
\alpha(1 \otimes (1,0)) &= (0,(1,1,0)) \\
\alpha(1 \otimes (0,1)) &= (0,(0,-1,1)) \\
\alpha(1 \otimes 1) &= (1 \otimes 1,(1,1,1))
\end{aligned}$$

Note that here we choose the generator of $\mathbb{Z}^2/\langle(1,1)\rangle$ is in the image of $(1,0)$. So $\ker(\alpha) \cong H_3(M_P)$ is trivial. Furthermore, the image of α is $\mathbb{Z} \oplus \mathbb{Z}^2 = \ker(\phi)$ so $H_2(M_P) = \mathbb{Z}$. Since β is an isomorphism, σ_2 and ϕ' are 0 and hence ϕ is surjective. So $H_2(M_P) = \mathbb{Z}$. Also, $H_1(M_P) =\ker(\gamma) = 0$ since γ is injective. Finally, $H_0(M_P) = (\mathbb{Z} \oplus \mathbb{Z})/\mathbb{Z} = \mathbb{Z}$ since ϕ'' is surjective.

In summary, we have

$$H_n(M_P) = \begin{cases} 0 & n = 1,3 \text{ and } n > 4 \\ \mathbb{Z} & n = 0,2,4 \end{cases}$$

which coincides with the homology groups of $\mathbb{CP}^2$. So this coincides with example 4.9.

4.4 Hirzebruch surface as a quotient space

In the last section we calculate the homology of the toric variety over a triangle. We would like to generalise this method to more applicable examples. In this section we look at a famous example of toric varieties from topological point of view.

The Hizebruch surface F^n of rank n is the toric variety constructed from the 2-dimensional polytope P with normal fan spanned by $l_1 = (1,0)$, $l_2 = (0,1)$, $l_3 = (-1,n)$ and $l_4 = (0,-1)$ as figure 12 shows. As an example, we take rank $n = 2$ and calculate its homology.

Again, we divide P in five parts as before so $F^2 = M_{P(5)} \cup M_{P(5)'}$ where $M_{P(5)} = p(5) \times T^2$

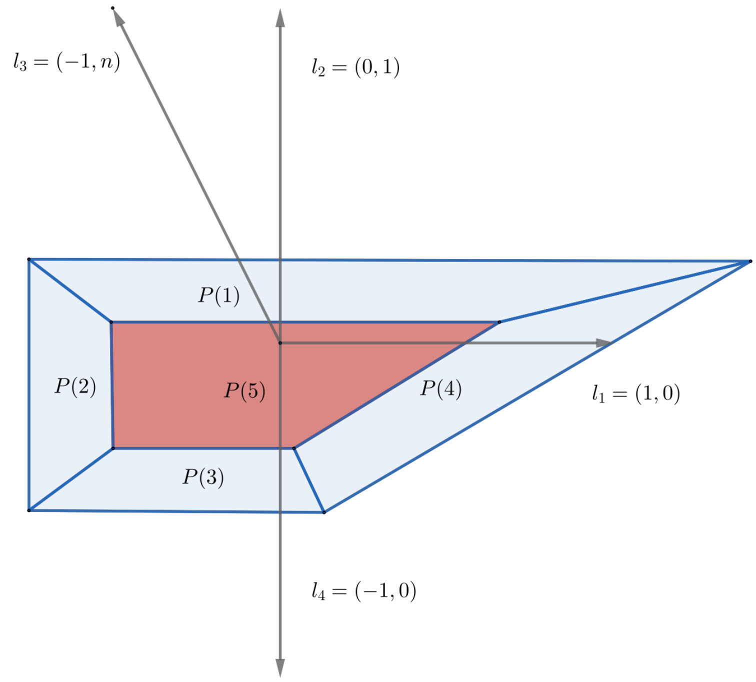


Figure 12: The Hirzebruch surface

and l_1, l_2, l_3 and l_4 projects to 4 subtori on four sides respectively. Gathering the inputs of the Mayer-Vietoris sequence, we see that the intersection is still $S^1 \times T^2$ and $M_{P(5)}$ is still homotopy equivalent to T^2. The only difference we get is that $M_{P(5)'}$ is homotopy equivalent to $S^1 \vee S^2 \vee S^2 \vee S^2 \vee S^2$ with four copies of S^2 rather than 3. So we get a very similar Mayer-Vietoris sequence:

$$\begin{array}{ccccccc}
0 & \longrightarrow & 0 & \longrightarrow & H_4(M_P) & & \\
\mathbb{Z} & \longrightarrow & 0 \oplus 0 & \longrightarrow & H_3(M_P) & & \\
\mathbb{Z}^3 & \xrightarrow{\alpha} & \overset{\sigma_3}{\mathbb{Z} \oplus \mathbb{Z}^4} & \xrightarrow{\phi} & H_2(M_P) & & \\
\mathbb{Z}^3 & \xrightarrow{\beta} & \overset{\sigma_2}{\mathbb{Z}^2 \oplus \mathbb{Z}} & \xrightarrow{\phi'} & H_1(M_P) & & \\
\mathbb{Z} & \xrightarrow{\gamma} & \overset{\sigma_1}{\mathbb{Z} \oplus \mathbb{Z}} & \xrightarrow{\phi''} & H_0(M_P) & \longrightarrow & 0
\end{array}$$

At the level of 0 and 1-homology, the maps γ and β remains unchanged. It follows that $H_n(M_P)$ is the same as in example 4.10. Also, the first factor of α is also an isomorphism. The only thing we need to figure out is the second factor of α. Denote the generators of $H_2(S^1 \times T^2))$ by g, g_1 and g_2 just as before. g is still sent to $(g, (1, 1, 1, 1))$ just like before.

Now recall that for each pieces $P(1), P(2)$, $P(3)$ and $P(4)$, we have the projection map for $i = 1, 2, 3, 4$:

$$T^2 \to T^2/T(i) \cong S^1$$

which induces a map of homology

$$\mathbb{Z} \oplus \mathbb{Z} \to \mathbb{Z}$$

with kernel generated by $(1, 0)$, $(0, 1)$, $(-1, 2)$ and $(0, -1)$ for $i = 1, 2, 3$ and 4 respectively. Therefore, for $i = 1$, the generator $g_1 = 1 \otimes (1, 0)$ in $M_\cap$ corresponds to 1 in M_{F_1}, 0 in M_{F_2}, 2 in M_{F_3}, and 0 in M_{F_4}. Similarly, g_2 corresponds to 0 in M_{F_1}, 1 in M_{F_2} and M_{F_3} and -1 in M_{F_4}. Hence by choosing appropriate generators, we have

$$\begin{aligned}
\alpha(1 \otimes (1, 0)) &= (0, (1, 0, 2, 0)) \\
\alpha(1 \otimes (0, 1)) &= (0, (0, 1, 1, -1)) \\
\alpha(1 \otimes 1) &= (1 \otimes 1, (1, 1, 1, 1))
\end{aligned}$$

So the image of α is still a rank 3 summand. So we have $H_2(F^2 = M_P) = \mathbb{Z}^5/\mathbb{Z}^2 = \mathbb{Z}^3$. However, in the above analysis, the rank of F^n only affect the coordinate $(-1, n)$ so if we do the same calculation for F^n, we only need to change 2 to n for any $n \geq 1$ without changing the result. Therefore, the homology groups does not depend on the rank:

$$H_n(F^m) = \begin{cases} 0 & n = 1, 3 \text{ and } n > 4 \\ \mathbb{Z} & n = 0, 4 \\ \mathbb{Z}^2 & n = 2 \end{cases}$$

for all $m \geq 2$.

Actually, this generalises to all toric varieties constructed from the normal fan of an integral polytope P in $\mathbb{R}^2$. So we can in principle perform the same calculation to any polygon in $\mathbb{R}^2$. The map α will always be injective hence the second homology group will be free with rank the number of edges of $P - 2$.

4.5 Outlook

One would ask the question: what about higher dimensions? Following the same construction as in section 4.2 and 4.3, we consider for example the unit cube P in $\mathbb{R}^3$:

$$M_P = P \times T^3/\sim$$

In each face of P, T^3 projects to a subtorus T^2 and in each edge a subtorus $T = S^1$. What we can do is to decompose the cube by setting a smaller cube C inside the interior of P. Then M_C is homotopy equivalent to T^3. Denote the complement of C by C'. We need to figure out the homology of $M_{C'}$ which is homotopy equivalent to $M_{\partial P}$. Denote the 1-skeleton by $P^{(1)}$. Then $M_{P(1)}$ is $P^{(1)}$ with each edge replaced by a copy of S^2 so we can figure out its homology by lemma 4.12. By projecting the faces of C to faces of P naturally, we have ∂P is the union of faces of C and a neighbourhood of $P^{(1)}$ which is homotopy equivalent to $P^{(1)}$. Therefore, by applying the Mayer-Vietoris sequence to we can calculate the homology of $M_{C'}$.

So we have the homology groups of M_C and $M_{C'}$, and we can calculate the homology of $M_C \cap M_{C'} = S^2 \times T^3$ via Kunneth formula. By applying the Mayer-Vietoris sequence again, we can finally calculate the homology groups of M_P as we did for 2-dimensional cases, although the analysis of maps in the sequences is more tedious to figure out.

Of course this generalises to all polytopes in $\mathbb{R}^3$. One would expect that higher dimensional cases can be calculated using the same method, but the amount of calculations may indicate that this is not effective and sometimes not feasible. Hence one would expect that more advanced tools are needed, for example the spectral sequences.

5 Conclusion

In conclusion, we discussed in section 2 that a convex polytope can be described in two equivalent ways: finite intersection of closed half-spaces(H-polytope) or the convex hull of finite many points(V-polytope). We also saw that for any given H-polytope polytope, there is a one-to-one correspondence between the facets and the normal vectors of defining half-spaces.

Next. we defined the toric varieties of a polytope as a topological space. Using the Mayer-Vietories sequences, we calculated that the toric variety M_P of a triangle P in $\mathbb{R}^2$ has homology groups

$$H_n(M_P) = \begin{cases} 0 & n = 1, 3 \text{ and } n > 4 \\ \mathbb{Z} & n = 0, 2, 4 \end{cases}$$

and which agrees with the geometric fact that it is homeomorphic to $\mathbb{CP}^2$ as a toric variety. As a preparation, we proved that a finite graph with $|E|$ edges and $|V$ vertices is homotopy equivalent to a wedge sum of $n = |E| - |V| + 1$ cycles S^1. Furthermore, if we replace each edge with a copy of S^2, the resulting space is homotopy equivalent to a wedge sum of n circles and $|E$ copies of S^2.

We also generalised the calculation to see that the homology groups of the Hirzebruch surfaces F^m does not depends on the rank:

$$H_n(F^m) = \begin{cases} 0 & n = 1, 3 \text{ and } n > 4 \\ \mathbb{Z} & n = 0, 4 \\ \mathbb{Z}^2 & n = 2 \end{cases}$$

Finally, we discussed the possibility of performing similar calculations in higher dimensions and concluded that this is feasible but may not be effective in dimension 3.

In summary, there are four main directions for further study:

(1) The Mayer-Vietoris sequence is not sufficient for calculating homology of toric varieties, especially in higher dimensions. So if time permitted, it would be reasonable to have introduced spectral sequences as an essential tool which hopefully simplifies the calculation by taking "approximations". Possible references are: [14] and [15].

(2) The current examples of convex polytopes we have are mainly in $\mathbb{R}^2$ and $\mathbb{R}^3$. One possible path is to find ways of "visualizing" polytopes in higher dimensions and this will help us manipulate and analyse polytopes in $\mathbb{R}^n$ with concrete, geometric intuitions. For example, Gale diagrams and Schlegel diagrams. Possible reference is [8].

(3) Convex polytopes can be associated with many combinatorial attributes, for example, the f-vectors and h-vectors. And the combinatorial information can be connected with the homology/cohomology of toric varieties and quasitoric manifolds(which we don't have time to touch on). Possible future study would be finding out the relevant theorems on both combinatorics and topology of toric varieties and quasitoric manifolds defined on convex polytopes. [3] and [4] are still very good sources.

(4) Toric geometry has become a separate branch of mathematics. In section 4.1 we briefly discussed the algebraic construction of toric varieties but there are more interesting stuff in this subject. For example, one could study divisors on toric varieties and hence define line bundles. In particular, the Hirzebruch surfaces can be described algebraically in the language of line bundles. Typical references includes [1] and [19].

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